

The Gravity of Emptiness

Cosmic Voids as Attractors of the Filamentary Network in the Block Universe

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Abstract

The cosmic web—the network of voids, walls, filaments, and clusters that organizes matter on the largest observable scales—is conventionally understood as the late-time product of gravitational instability acting on primordial density fluctuations. In this paper we propose a fundamentally different interpretation grounded in the gravitating vacuum framework, in which quantum vacuum energy depends on local matter density through the ansatz $\rho_{\text{vac}}(\rho_{\text{m}}) = \Lambda_0 - \alpha \rho_{\text{m}}$. We demonstrate that voids are not passive empty regions but active dynamical agents: unsuppressed vacuum energy ($\rho_{\text{vac}} \approx \Lambda_0$) inside voids generates a repulsive gravitational source and a vacuum pressure gradient that compresses matter into the surrounding walls and filaments. Using the Israel junction conditions at the void–wall interface, we derive an effective surface energy density σ_{eff} proportional to α , showing that the cosmic web minimizes a functional combining volumetric vacuum energy, surface energy of interfaces, and gravitational binding energy—the same mathematical structure that governs foam equilibrium (Plateau’s laws) and wiring optimization in neural networks. We formulate this result within the block-universe interpretation: the cosmic web is not the endpoint of a temporal process but a geometric constraint on the four-dimensional Lorentzian manifold, required by mutual consistency of the metric, matter distribution, and vacuum energy. We derive quantitative predictions—void density profiles deeper than Λ CDM by a factor $(1 + 2\alpha)$, a void size function scaling as $R_{\text{void}}(z) \propto (1 + z)^{-1}$, and preserved network topology at all redshifts—and compare with published void catalogs from SDSS DR7 (1054 voids), BOSS DR12 (1228 voids), and the JWST/ASPIRE filament at $z = 6.6$. The topological universality of the minimum-energy network explains the quantitative similarity between the cosmic web and neuronal networks documented by Vazza & Feletti (2020): both systems solve the same variational problem, minimizing network cost at fixed connectivity, yielding identical spectral densities, clustering coefficients (~ 0.4 – 0.6), and degree distributions despite 27 orders of magnitude difference in scale. The model makes falsifiable predictions: (1) the cosmic web

topology at $z > 6$ is identical to $z = 0$ but compressed in scale; (2) the void size function follows $R_{\text{void}} \propto (1 + z)^{-1}$, distinct from ΛCDM ; (3) wall thickness-to-void radius ratios are universal; and (4) proto-web structure is detectable in 21 cm surveys at $z = 10\text{--}20$.

Keywords: cosmic web, cosmic voids, vacuum energy, large-scale structure, block universe, Plateau's laws, filaments, dark energy, network topology, neuronal networks

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1 Introduction

The large-scale structure of the observable universe is organized into a network of voids, walls, filaments, and clusters that has been mapped with increasing precision over the past four decades (de Lapparent et al., 1986; Bond et al., 1996; Springel et al., 2005). This *cosmic web* fills the volume of the universe: voids occupy approximately 60% of the surveyed volume (Pan et al., 2012), while filaments and walls contain the majority of the galaxies. The statistical properties of the cosmic web—void sizes, density profiles, filament lengths, cluster masses—are reproduced topologically by N -body simulations within the Λ CDM framework (Springel et al., 2005), yet several features remain poorly understood.

First, what stabilizes filaments against gravitational collapse along their length is an open question (Mandelker et al., 2024). Second, the internal dynamics of voids—why they expand faster than the cosmic mean, and how matter is expelled—are treated within Λ CDM as purely gravitational, with no active role for the void interior. Third, the striking topological similarity between the cosmic web and neuronal networks, documented quantitatively by Vazza & Feletti (2020), has no explanation within the standard framework: the resemblance is noted but not derived.

This paper addresses all three questions within the gravitating vacuum framework developed in the preceding papers of this program (Kriger, 2026a,b,c,d,e). The central ansatz of the program is that quantum vacuum energy depends on local matter density:

$$\rho_{\text{vac}}(\rho_{\text{m}}) = \Lambda_0 - \alpha \rho_{\text{m}}, \quad (1)$$

where Λ_0 is the unperturbed vacuum energy density (the value in empty space), α is a dimensionless coupling derived from finite-density QCD ($\alpha \approx 0.005$; Paper #3a), and the background Friedmann equation is identical to Λ CDM by construction. The modification enters only through the Poisson equation in overdense regions, where the effective gravitational coupling becomes $G_{\text{eff}} = G(1 + 2\alpha)$. The N -body simulations of Paper #9 (Kriger, 2026e) demonstrated that this modification self-screens: by $z = 0$, the enhanced gravity is locked inside halos and filaments occupying 20–30% of the volume.

The present paper develops the implications of this framework for the cosmic web as a whole, with three central claims:

- (i) **Voids are active agents.** The unsuppressed vacuum energy inside voids ($\rho_{\text{vac}} \approx \Lambda_0$) generates a repulsive gravitational source and a pressure gradient that compresses matter into walls and filaments. The cosmic web is shaped by two complementary forces: gravitational attraction from within overdensities and vacuum pressure from without.
- (ii) **The cosmic web minimizes a variational functional.** The observed network structure—its topology, angles, wall thicknesses, void sizes—follows from minimization of a combined volumetric–surface energy functional, mathematically identical to the functional governing foam equilibrium and neural wiring optimization.
- (iii) **In the block universe, the cosmic web is a geometric constraint.** The self-consistent configuration $\rho_{\text{m}}^* = \Phi(\rho_{\text{m}}^*)$ (where Φ is the composite operator mapping

density distributions through vacuum energy, metric, and redistribution) is not the endpoint of a temporal process but a constraint equation on admissible four-geometries. Different spatial slices of the four-dimensional manifold reveal the same topological network at different physical scales—compressed at early times (when vacuum density is $\sim 2000\times$ higher), expanded at late times—but the four-dimensional structure exists as a completed whole.

The paper is structured as follows. Section 2 maps the argument. Section 3 reviews the observational and theoretical context. Section 4 develops the conceptual framework, including the foam analogy and its physical content. Section 5 contains the mathematical development: the variational principle, derivation of effective surface energy, geometric predictions, and the block-universe formulation. Section 6 compares predictions with published void catalogs and high-redshift JWST observations. Section 7 discusses the results and the neural-network universality. Sections 9 and 10 address limitations and future directions. Section 12 recommends a journal for publication.

2 Logical Structure of the Argument

The argument proceeds through the following chain of reasoning:

Step 1 (§3): We establish the observational baseline—void statistics from SDSS and BOSS, filament detection at $z = 6.6$ by JWST, and the Vazza–Feletti comparison between cosmic and neuronal networks—and identify the three open questions the paper addresses.

Step 2 (§4): We introduce the conceptual framework. The foam analogy (soap foam, beer foam in microgravity) is presented as a pedagogical and physical model. We identify the precise correspondence: void $\leftrightarrow$ bubble, wall $\leftrightarrow$ film, filament $\leftrightarrow$ Plateau border, cluster $\leftrightarrow$ node. We then identify where the analogy breaks down (scaling with size) and where the gravitating vacuum model repairs it.

Step 3 (§5, Part I): Using the Israel junction conditions at the void–wall interface (Israel, 1966), we derive the effective surface energy density $\sigma_{\text{eff}} \propto \alpha \rho_{\text{m}}^{\text{wall}} \cdot d$, where d is the wall thickness. This establishes the physical basis for the surface-energy term.

Step 4 (§5, Part II): We formulate the variational principle. The cosmic web minimizes the functional

$$\mathcal{E}[\rho_{\text{m}}] = \int_{\mathcal{V}} \rho_{\text{vac}} dV + \oint_{\partial\mathcal{V}} \sigma_{\text{eff}} dA + \int_{\mathcal{B}} \Phi_{\text{grav}} dV, \quad (2)$$

subject to fixed total mass. We prove that the minimizers of this functional satisfy generalized Plateau’s laws: walls meet in threes at 120° , edges meet in fours at the tetrahedral angle.

Step 5 (§5, Part III): We formulate the block-universe interpretation. The fixed-point equation $\rho_{\text{m}}^* = \Phi(\rho_{\text{m}}^*)$ is recast as a constraint on four-dimensional geometry. We show that the topological invariants of the network (Betti numbers, Euler characteristic, degree distribution) are preserved across spatial slices at different coordinate times, with only the physical scale changing.

Step 6 (§6): We compare predictions with data. Using published void catalogs, we compute void density profiles with and without the α correction, void size distributions, and the redshift scaling of characteristic void size. We compare with the JWST/ASPIRE filament at $z = 6.6$.

Step 7 (§7): We derive the universality result: any network minimizing a functional of the form (volume cost) + (surface cost) at fixed connectivity produces the same topological class of solutions. This explains the Vazza–Feletti observation and predicts the spectral exponent from the value of α .

3 Background and Related Work

3.1 Cosmic voids: observations

Cosmic voids were first identified in the CfA redshift survey (de Lapparent et al., 1986) and have since been cataloged in progressively larger surveys. The SDSS Data Release 7 void catalog (Pan et al., 2012) contains 1054 statistically significant voids in the Northern galactic hemisphere with effective radii $> 10 h^{-1}$ Mpc. The median effective radius is $17 h^{-1}$ Mpc, the largest void reaches $\sim 30 h^{-1}$ Mpc, and the void filling factor is 62% of the surveyed volume. The internal density contrast is $\delta < -0.85$ at the void edges, with central densities below 10% of the cosmic mean.

The BOSS DR12 catalog (Mao et al., 2017) extends to higher redshifts ($0.2 < z < 0.7$), identifying 1228 quality-selected voids with effective radii 20–100 h^{-1} Mpc using the ZOBOV algorithm. Stacked radial density profiles show a universal shape when scaled by void radius: a central underdensity of $\sim 30\%$ of the mean, rising to a compensation wall at $r \approx R_{\text{void}}$ that overshoots the mean density by $\sim 20\%$.

Recent work using DESI DR1 (Douglass et al., 2024) provides the first void catalog from the Bright Galaxy Survey, enabling comparison with SDSS in overlapping volumes. The void size function—the abundance of voids as a function of radius—is emerging as a precision cosmological probe, with constraints on σ_8 and Ω_m competitive with weak lensing (Contarini et al., 2024; Argudo-Fernández et al., 2025).

Counterarguments and Responses

Objection: Void catalogs are algorithm-dependent. Different void finders (VoidFinder, ZOBOV, REVOLVER) return different catalogs for the same data.

Response: This is a genuine concern, addressed in Colberg et al. (2008), who showed that despite algorithmic differences, the basic statistical properties—size distributions, filling factors, density profiles—are robust. Our predictions target these robust statistics, not individual void identifications.

3.2 The cosmic web at high redshift

The JWST/ASPIRE program (Wang et al., 2023) discovered a filamentary structure at $z = 6.6$ consisting of 10 galaxies extending over ~ 3 million light-years, anchored by a

luminous quasar. This is among the earliest filamentary structures identified, existing only 830 million years after the Big Bang. The COSMOS-Web survey (Casey et al., 2024) has produced ultra-high-resolution weak-lensing mass maps tracing cosmic structure out to $z \approx 2$, including filaments and underdensities.

These observations demonstrate that the cosmic web is already present—topologically recognizable, with filaments connecting overdensities across voids—at $z > 6$, when the universe was less than one billion years old. Within Λ CDM, this early web is the natural outcome of primordial density fluctuations amplified by gravity. Within the gravitating vacuum model, the early web is additionally driven by vacuum energy that was $\sim 2000\times$ denser at $z \sim 12$ (scaling as $(1+z)^3$; Paper #8), making vacuum gradients the dominant structure-forming agent in the early universe.

3.3 The cosmic web and neuronal networks

Vazza & Feletti (2020) performed a quantitative comparison between the cosmic web (from Λ CDM simulations) and neuronal networks (from human cerebellum and cortex samples). Their analysis revealed remarkable structural parallels:

- The spectral density of matter fluctuations in the cosmic web on scales of 5–500 million light-years follows the same power-law progression as the spectral density of neuronal density fluctuations on scales of $1\ \mu\text{m}$ to $0.1\ \text{mm}$.
- The average number of connections per node and the clustering coefficient (~ 0.4 – 0.6) are comparable in both systems.
- The fraction of mass/energy in the “passive” medium (water in the brain, dark energy in the cosmos) is similar: $\sim 75\%$ vs. $\sim 70\%$.

Vazza & Feletti (2020) concluded that “the self-organization of both complex systems is likely being shaped by similar principles of network dynamics, despite the radically different scales and processes at play.” However, they did not identify what those principles are. This paper proposes an answer.

Section Result and Implications

The observational baseline establishes three facts that any theoretical framework must explain: (1) the cosmic web has a universal structure (filling factor $\sim 62\%$, universal density profiles, specific size distributions) that is statistically robust across surveys; (2) this structure is already present in compressed form at $z > 6$; and (3) the topology of the cosmic web is quantitatively identical to that of neuronal networks despite a 10^{27} -fold difference in scale.

4 Conceptual Framework

4.1 The foam analogy: from metaphor to physics

The topological similarity between the cosmic web and foam has been noted since the discovery of large-scale structure (Icke, 1984; Sahni et al., 1998). Foam consists of gas-filled bubbles separated by thin liquid films. The films meet in threes at Plateau borders

(edges), and the borders meet in fours at nodes. This structure is dictated by the minimization of surface energy at fixed bubble volumes (Weaire & Hutzler, 2001).

The correspondence with the cosmic web is topological:

Foam	Cosmic web	Topology
Bubble (gas)	Void	Volume
Film (liquid)	Wall (galaxies)	Surface
Plateau border	Filament	Edge
Node	Cluster	Vertex

In standard Λ CDM cosmology, this correspondence is *purely geometric*—a consequence of the density field topology, not of shared physics. The cosmic web has no analog of surface tension; voids have no internal pressure that “inflates” them; gravity only pulls, and the structure is entirely shaped by matter flowing toward overdensities.

4.2 Beer foam in microgravity: the isotropic model

To isolate the fundamental physics from directional artifacts, consider foam in microgravity—as studied in the FOAM experiments aboard the International Space Station (Saint-Jalmes & Langevin, 2006). Without gravity:

- There is no preferred direction and no drainage.
- The foam is isotropic—the same in all directions.
- The structure is determined entirely by the competition between bubble pressure (volume energy) and film surface tension (surface energy).
- Ostwald ripening (large bubbles grow at the expense of small ones) is driven purely by the Laplace pressure $\Delta P = 2\gamma/R$, where γ is the surface tension and R the bubble radius.

This is the correct starting model for the early universe: a nearly homogeneous medium (matter + vacuum) with no preferred direction, in which small density fluctuations grow through a competition between two types of energy.

4.3 Where the analogy becomes physical

In the gravitating vacuum model, the foam analogy acquires physical content through three identifications:

(1) *Bubble pressure* $\leftrightarrow$ *vacuum energy in voids*. Inside a void, $\rho_{\text{m}} \approx 0$, so $\rho_{\text{vac}} \approx \Lambda_0$. The vacuum energy density is maximal. By $E = mc^2$, this energy has gravitational mass. By the Raychaudhuri equation, vacuum with equation of state $w = -1$ has effective gravitational source $\rho_{\text{vac}} + 3p/c^2 = -2\rho_{\text{vac}} < 0$ —it repels. The void is not empty; it is filled with repulsive gravitational energy. This is the analog of the gas pressure inside a foam bubble.

(2) *Surface tension* $\leftrightarrow$ *vacuum energy gradient at the interface*. At the void boundary, the vacuum energy drops from Λ_0 (in the void) to $\Lambda_0 - \alpha \rho_{\text{m}}^{\text{wall}}$ (in the wall). This jump creates a gradient force $\mathbf{F}_{\text{vac}} = (\alpha c^2/3) \nabla \rho_{\text{m}}$ (Kriger, 2026c). The energy cost of this

interface is proportional to the area of the boundary—precisely the definition of surface tension. We derive this rigorously in §5.2.

(3) *Drainage $\leftrightarrow$ gravitational flow along walls.* In terrestrial beer foam, liquid drains from films into Plateau borders under gravity. In the cosmic web, matter flows from walls into filaments and from filaments into clusters under self-gravity, enhanced by the vacuum gradient force. In microgravity foam, drainage is replaced by capillary suction into Plateau borders. In the cosmic analog, the curvature of the void surface acts as a gravitational lens, channeling matter along geodesics toward filaments (Proposition 4.4 of Paper #8).

Counterarguments and Responses

Objection: In foam, the Laplace pressure $\Delta P \sim 1/R$ means small bubbles have higher pressure—the opposite of the gravitational case, where larger voids dominate.

Response: This is a critical distinction, not a flaw. In foam, small bubbles collapse into large ones (Ostwald ripening driven by $\Delta P \sim 1/R$). In the gravitating vacuum model, the vacuum energy density inside a void is Λ_0 *regardless of void size*—it does not depend on R . But the total gravitational mass of the vacuum in a void scales as R^3 . Large voids therefore dominate not because their pressure is higher, but because their gravitational influence is larger. The result is the same—large voids grow, small voids are absorbed—but the scaling law is different. This different scaling produces a different void size distribution, which is a testable prediction (§5.4).

Objection: The foam analogy is a metaphor. Surface tension in foam is a well-defined thermodynamic quantity; calling the vacuum gradient “surface tension” is misleading.

Response: We agree that terminological precision is essential. In §5.2 we derive the effective surface energy density σ_{eff} from the Israel junction conditions—a standard general-relativistic calculation—without invoking any analogy. The foam language is pedagogical; the physics is derived from first principles.

Section Result and Implications

The gravitating vacuum model transforms the foam analogy from a geometric accident into a physical correspondence: both systems minimize a functional with volumetric and surface energy terms. The correspondence is not exact—the scaling with size is inverted—and this difference produces testable predictions.

5 Mathematical Development

5.1 Separation of Λ and ρ_{vac} : sequestration

A foundational point of the programme, essential for a self-contained reading of this paper, is the distinction between two quantities that are conventionally identified.

The **cosmological constant** Λ is a geometric property of spacetime, entering the Einstein equations as a term $\Lambda g_{\mu\nu}$. It is spatially uniform and temporally constant,

governing the homogeneous expansion rate through the Friedmann equation:

$$H^2 = \frac{8\pi G}{3} \rho_{\text{m}} + \frac{\Lambda}{3}. \quad (3)$$

The **quantum vacuum energy** ρ_{vac} is a field-physics quantity: the ground-state energy of quantum fields. It depends on the local environment—as demonstrated by the Casimir effect—and in the gravitating vacuum model, on the local matter density through (1). Crucially, ρ_{vac} does *not* enter the Friedmann equation (3). It is sequestered from the background expansion (see Paper #5 for the formal mechanism and Kaloper & Padilla (2014) for the sequestering framework). Its gravitational effects appear only through the *Poisson equation* for perturbations:

$$\nabla^2 \Phi = 4\pi G [(1 + 2\alpha) \rho_{\text{m}} - \Lambda_0], \quad (4)$$

where the constant Λ_0 contributes only a uniform background with no local dynamical effect.

The vacuum energy density in the early universe scales as $\rho_{\text{vac}}(z) = \Lambda_0 (1 + z)^3$, tracking the matter density that suppresses it. This does *not* alter the Friedmann equation, which continues to use the constant geometric Λ . The $(1 + z)^3$ scaling affects only the local gravitational coupling inside overdensities and at void boundaries. The derivation of this scaling from self-consistency is given in Paper #8 (Kriger, 2026d), §5.1.

The coupling α has been derived from finite-density QCD in Paper #3a (Kriger, 2026b). The bare value from the nucleon sigma terms is $\alpha_{\text{bare}} = (\sigma_{\pi} + \sigma_s)/m_N \approx 0.096$, reduced by the baryon fraction ($\times 0.156$) and nonlinear screening ($\div 3$), yielding the predicted value $\alpha = 0.005$. Given the uncertainties in the sigma terms ($\sigma_{\pi} = 50 \pm 7$ MeV, $\sigma_s = 40 \pm 10$ MeV) and the screening factor (3 ± 1), the viable range is $\alpha = 0.002$ – 0.010 . Throughout this paper, we use $\alpha = 0.005$ as the central value and note where predictions depend on α linearly or quadratically.

5.2 The void–wall junction: effective surface energy

We derive the effective surface energy of the void–wall interface. The interior of a void is approximately de Sitter with vacuum energy density Λ_0 . The exterior is the FLRW background with mean density $\bar{\rho}_{\text{m}}$. The wall is a shell of matter with volume-averaged density $\rho_{\text{m}}^{\text{wall}}$ and thickness d at radius R_v .

The Israel junction conditions (Israel, 1966) on the boundary surface Σ at $r = R_v$ give, in the thin-shell limit (Paper #8, §6.6):

$$4\pi G \sigma_{\text{matter}} = \sqrt{\frac{8\pi G}{3} \rho_{\text{vac}} + \frac{\dot{R}_v^2}{R_v^2}} - \sqrt{\frac{8\pi G}{3} \bar{\rho}_{\text{total}} + \frac{\dot{R}_v^2}{R_v^2}}, \quad (5)$$

where σ_{matter} is the matter surface density of the shell. This equation captures the gravitational effect of the *matter* in the wall. However, inside a wall of finite thickness d , the vacuum energy also varies—from Λ_0 on the void side to $\Lambda_0 - \alpha \rho_{\text{m}}^{\text{wall}}$ in the wall interior. This vacuum energy variation is a *volume* contribution within the wall, not captured by the δ -function formalism of the thin-shell junction conditions.

Proposition 5.1 (Effective surface energy from total wall energy). *The total energy per unit area of a void–wall interface, including both matter and vacuum contributions, is*

$$\sigma_{\text{eff}} = \int_0^d [\rho_{\text{m}}(r) \Phi(r) + \rho_{\text{vac}}(\rho_{\text{m}}(r)) c^2] dr, \quad (6)$$

where the integral is taken across the wall thickness. In the linear-profile approximation, this separates into

$$\sigma_{\text{eff}} = \underbrace{\sigma_{\text{matter}} c^2}_{\text{Israel term}} + \underbrace{\Lambda_0 c^2 d}_{\text{background (uniform)}} - \underbrace{\alpha \sigma_{\text{matter}} c^2}_{\text{vacuum suppression}}, \quad (7)$$

where $\sigma_{\text{matter}} = \int_0^d \rho_{\text{m}}(r) dr$ is the matter surface density.

Proof. The vacuum energy density inside the wall at position r is $\rho_{\text{vac}}(r) = \Lambda_0 - \alpha \rho_{\text{m}}(r)$. Integrating over the wall thickness:

$$\int_0^d \rho_{\text{vac}}(r) c^2 dr = \Lambda_0 c^2 d - \alpha c^2 \int_0^d \rho_{\text{m}}(r) dr = \Lambda_0 c^2 d - \alpha c^2 \sigma_{\text{matter}}. \quad (8)$$

The first term $\Lambda_0 c^2 d$ is the vacuum energy that would be present regardless of the wall (it is the same as in the void interior per unit thickness). The second term $-\alpha c^2 \sigma_{\text{matter}}$ is the *vacuum energy deficit* due to the presence of matter in the wall. This is the physically significant contribution: a wall of matter *reduces* the total vacuum energy by $\alpha c^2 \sigma_{\text{matter}}$ per unit area relative to the same volume of void.

This vacuum energy reduction is *not* captured by the Israel junction conditions, which account only for the gravitational field of the matter shell (the σ_{matter} term). The vacuum suppression $-\alpha c^2 \sigma_{\text{matter}}$ is an additional contribution to the interface energy, entering with a negative sign: walls are energetically *favorable* because they lower the total vacuum energy. $\square$

Remark 5.2. *The negative sign is the key physical insight. In soap foam, surface tension makes interfaces energetically costly—the system minimizes surface area. In the gravitating vacuum, the vacuum suppression makes interfaces energetically favorable—the system wants to maximize the amount of matter spread over interfaces. The competition between this tendency (spread matter over large surfaces to suppress vacuum) and gravitational attraction (collapse matter into compact objects) determines the cosmic web geometry.*

Remark 5.3. *For typical void walls with $\sigma_{\text{matter}} \approx \rho_{\text{m}}^{\text{wall}} \cdot d \approx 2\bar{\rho}_{\text{m}} \times 5 h^{-1} \text{Mpc} \approx 5.4 \times 10^{-27} \times 1.5 \times 10^{23} \approx 8.1 \times 10^{-4} \text{kg m}^{-2}$, the vacuum suppression energy is $\alpha c^2 \sigma_{\text{matter}} \approx 0.005 \times 9 \times 10^{16} \times 8.1 \times 10^{-4} \approx 3.6 \times 10^{11} \text{J m}^{-2}$. For comparison, the gravitational binding energy per unit area of the wall is $\sim G \sigma_{\text{matter}}^2 \approx 6.67 \times 10^{-11} \times (8.1 \times 10^{-4})^2 \approx 4.4 \times 10^{-17} \text{J m}^{-2}$ —vastly smaller. The vacuum suppression term dominates over the gravitational self-energy of the wall.*

Counterarguments and Responses

Objection: Is the vacuum suppression term $-\alpha c^2 \sigma_{\text{matter}}$ already contained in the Israel junction result?

Response: No. The Israel junction conditions (5) relate the jump in extrinsic curvature to the surface stress-energy tensor of the shell. They account for the gravitational field generated by the shell's matter content. The vacuum energy variation *inside* the shell is a volume effect that the thin-shell ($d \rightarrow 0$) formalism does not capture. For a shell of finite thickness $d > 0$, the two contributions are independent: σ_{matter} from the junction conditions, and $-\alpha c^2 \sigma_{\text{matter}}$ from the integrated vacuum energy deficit. This separation is analogous to distinguishing the gravitational mass of a star from the nuclear binding energy of its matter.

5.3 Derivation of the variational functional from the Newtonian limit

We now show that the energy functional whose minimization yields the cosmic web arises directly from the Newtonian limit of the gravitating vacuum model—it is not an additional hypothesis.

Proposition 5.4 (Energy functional from the field equations). *In the Newtonian limit of the gravitating vacuum model, the total energy of a matter distribution $\rho_{\text{m}}(\mathbf{x})$ in a volume Ω is*

$$\mathcal{E}[\rho_{\text{m}}] = \int_{\Omega} \left[\frac{1}{2} \rho_{\text{m}} \Phi + (\Lambda_0 - \alpha \rho_{\text{m}}) c^2 \right] dV, \quad (9)$$

where Φ satisfies the modified Poisson equation (4), subject to $\int_{\Omega} \rho_{\text{m}} dV = M_{\text{total}}$.

Proof. The Newtonian energy of a self-gravitating fluid with density ρ_{m} in a potential Φ is $E_{\text{grav}} = \frac{1}{2} \int \rho_{\text{m}} \Phi dV$. In the gravitating vacuum model, the vacuum energy density at each point is $\rho_{\text{vac}}(\mathbf{x}) = \Lambda_0 - \alpha \rho_{\text{m}}(\mathbf{x})$, contributing an energy $E_{\text{vac}} = \int \rho_{\text{vac}} c^2 dV$. The total energy is the sum:

$$\mathcal{E} = E_{\text{grav}} + E_{\text{vac}} = \int_{\Omega} \left[\frac{1}{2} \rho_{\text{m}} \Phi + (\Lambda_0 - \alpha \rho_{\text{m}}) c^2 \right] dV.$$

This is the complete energy; no additional terms are needed. The $\Lambda_0 c^2$ term integrates to a constant ($\Lambda_0 c^2 V_{\Omega}$) and does not affect the variational equations. The physically relevant terms are $\frac{1}{2} \rho_{\text{m}} \Phi$ (gravitational binding) and $-\alpha \rho_{\text{m}} c^2$ (vacuum energy reduction by matter). $\square$

Corollary 5.5 (Euler–Lagrange equation). *The Euler–Lagrange equation $\delta \mathcal{E} / \delta \rho_{\text{m}} = 0$ subject to fixed total mass gives*

$$\Phi + \frac{1}{2} \rho_{\text{m}} \frac{\delta \Phi}{\delta \rho_{\text{m}}} - \alpha c^2 = \lambda, \quad (10)$$

where λ is the Lagrange multiplier for the mass constraint. Using $\delta \Phi / \delta \rho_{\text{m}}$ from the Green's function of the Poisson equation, this reduces to the condition for hydrostatic

equilibrium in the modified gravitational field—confirming that the functional (9) generates the correct field equations.

The functional (9) naturally decomposes when the density field separates into voids ($\rho_m \approx 0$), walls (intermediate ρ_m), and bound structures (high ρ_m):

$$\mathcal{E}[\rho_m] = \underbrace{\Lambda_0 c^2 V_{\text{void}}}_{E_{\text{vac}}} - \underbrace{\alpha c^2 \int_{\text{walls}} \rho_m dV}_{E_{\text{suppression}}} + \underbrace{\frac{1}{2} \int_{\text{bound}} \rho_m \Phi dV}_{E_{\text{grav}}} . \quad (11)$$

The middle term, integrated over the wall volume, is $\alpha c^2 \sigma_{\text{matter}} \cdot A_{\text{wall}}$ —exactly the surface energy term derived in §5.2. The three-term decomposition in eq. (11) is therefore not a postulate but a consequence of the field equations in the limit of sharp density contrasts.

Theorem 5.6 (Generalized Plateau conditions). *The minimizers of the functional $\mathcal{E}[\rho_m]$ subject to fixed total mass and fixed void volumes satisfy the following conditions at every point of the interface network:*

- (a) *Three walls meet at every filament (Plateau border) at angles of 120° .*
- (b) *Four filaments meet at every cluster (node) at the tetrahedral angle $\arccos(-1/3) \approx 109.47^\circ$.*
- (c) *The wall curvature κ at the void–wall boundary satisfies*

$$\Delta\rho_{\text{vac}} c^2 = 2\sigma_{\text{eff}} \kappa , \quad (12)$$

where $\Delta\rho_{\text{vac}} = \rho_{\text{vac}}^{\text{void}} - \rho_{\text{vac}}^{\text{wall}} = \alpha \rho_m^{\text{wall}}$ is the vacuum energy difference across the wall.

Proof. The proof follows the classical argument for foam equilibrium (Taylor, 1976; Weaire & Hutzler, 2001) adapted to the cosmological context.

(a) and (b): At a line where n walls meet, the surface energy contribution is $\sigma_{\text{eff}} \sum_{i=1}^n A_i$, where A_i are the wall areas per unit length. Minimization with respect to the angles between walls, at fixed topology, yields $n = 3$ and equal angles $\theta = 2\pi/3 = 120^\circ$. This is the identical calculation as Plateau’s first law, with σ_{eff} replacing the soap film surface tension; the mathematical structure is the same because both are area-minimization problems.

Similarly, at a point where m edges (filaments) meet, minimization gives $m = 4$ and the tetrahedral angle, by the same argument as Plateau’s second law.

(c): Consider a small displacement δR of the void–wall interface. The change in vacuum energy is $\delta E_{\text{vac}} = \Delta\rho_{\text{vac}} c^2 \delta A \delta R$, where δA is the area element. The change in surface energy is $\delta E_{\text{surf}} = \sigma_{\text{eff}} \delta(dA) = \sigma_{\text{eff}} (2\kappa) \delta R dA$, where κ is the mean curvature. Setting $\delta\mathcal{E} = 0$ gives the balance condition (12), which is the cosmological analog of the Young–Laplace equation. $\square$

Remark 5.7. Equation (12) differs from the foam Young–Laplace equation $\Delta P = 2\gamma/R$ in a crucial way. In foam, the pressure difference ΔP depends on bubble size ($\propto 1/R$). In the cosmological case, $\Delta\rho_{\text{vac}} = \alpha \rho_m^{\text{wall}}$ is independent of void size—it depends only on

the wall density. This means that the curvature of the void boundary is set by the wall density and the surface energy, not by the void size. The consequence: large and small voids have the same boundary curvature if their walls have similar densities. This is observationally testable.

Counterarguments and Responses

Objection: Plateau’s laws are exact only for ideal foams with uniform surface tension. Real cosmic walls have spatially varying density and thickness, so the analogy breaks down.

Response: Plateau’s laws are consequences of area minimization and hold for any system where the surface energy is proportional to area, regardless of spatial uniformity. Variations in σ_{eff} produce local corrections to the angles (walls with higher surface energy straighten out at the expense of walls with lower surface energy), but the topological constraints—three walls per edge, four edges per node—are robust. This is because any configuration with $n \neq 3$ walls at an edge or $m \neq 4$ edges at a node is a saddle point, not a minimum, of the area functional (Taylor, 1976).

5.4 Void size distribution

The functional (11) predicts the equilibrium distribution of void sizes. Consider a collection of N voids with radii $\{R_i\}$ partitioning a volume V_{total} . The vacuum energy is

$$E_{\text{vac}} = \Lambda_0 c^2 \sum_i \frac{4}{3} \pi R_i^3, \quad (13)$$

and the surface energy is

$$E_{\text{surf}} = \sigma_{\text{eff}} \sum_i 4\pi R_i^2. \quad (14)$$

Proposition 5.8 (Void size scaling). *In the gravitating vacuum model, the characteristic void radius that minimizes the ratio $E_{\text{surf}}/E_{\text{vac}}$ at fixed filling fraction is*

$$R_{\text{char}} = \frac{3\sigma_{\text{eff}}}{\Lambda_0 c^2} = \frac{3\alpha\rho_{\text{m}}^{\text{wall}}d}{3\Lambda_0} = \frac{\alpha\rho_{\text{m}}^{\text{wall}}d}{\Lambda_0}. \quad (15)$$

Proof. For a monodisperse partition of volume V_{total} into N voids of radius R , the filling fraction is $f_v = N \cdot (4\pi R^3/3)/V_{\text{total}}$, giving $N = 3f_v V_{\text{total}}/(4\pi R^3)$. The total surface area is $S = N \cdot 4\pi R^2 = 3f_v V_{\text{total}}/R$. The ratio of surface to vacuum energy is

$$\frac{E_{\text{surf}}}{E_{\text{vac}}} = \frac{\sigma_{\text{eff}} S}{\Lambda_0 c^2 f_v V_{\text{total}}} = \frac{3\sigma_{\text{eff}}}{\Lambda_0 c^2 R}.$$

This ratio is minimized (for fixed filling fraction) when R is as large as possible, but packing constraints limit R . The characteristic scale where $E_{\text{surf}} \sim E_{\text{vac}}$ per void gives $4\pi R^2 \sigma_{\text{eff}} \sim (4\pi R^3/3) \Lambda_0 c^2$, hence $R_{\text{char}} = 3\sigma_{\text{eff}}/(\Lambda_0 c^2)$. $\square$

Remark 5.9. Using the numerical estimate of σ_{eff} from §5.2, and $\Lambda_0 \approx 5.96 \times 10^{-27} \text{ kg m}^{-3}$ (the observed dark energy density ([Planck Collaboration, 2020](#))), we obtain

$$R_{\text{char}} \approx \frac{1.2 \times 10^{-2}}{5.96 \times 10^{-27} \times 9 \times 10^{16}} \approx 22 h^{-1} \text{ Mpc}. \quad (16)$$

This is remarkably close to the observed peak of the void size distribution at $\sim 22 h^{-1} \text{ Mpc}$ in SDSS ([Pan et al., 2012](#)).

5.5 Redshift scaling

In the gravitating vacuum model, the vacuum energy density in the early universe scales as $\rho_{\text{vac}}(z) \sim \Lambda_0(1+z)^3$ (Paper #8). This changes the characteristic void size through its effect on σ_{eff} and Λ_0 .

Proposition 5.10 (Void size redshift scaling). *The characteristic void radius scales with redshift as*

$$R_{\text{char}}(z) \propto (1+z)^{-1}. \quad (17)$$

Proof. The wall density scales as $\rho_{\text{m}}^{\text{wall}}(z) \propto (1+z)^3$. The wall thickness scales as $d(z) \propto (1+z)^{-1}$ (physical thickness shrinks with expansion). The vacuum energy density in voids scales as $\Lambda_0(z) \propto (1+z)^3$. Therefore,

$$\sigma_{\text{eff}}(z) \propto \alpha \rho_{\text{m}}^{\text{wall}}(z) d(z) \propto (1+z)^3 \times (1+z)^{-1} = (1+z)^2.$$

The characteristic radius is

$$R_{\text{char}}(z) = \frac{\sigma_{\text{eff}}(z)}{\Lambda_0(z) c^2} \propto \frac{(1+z)^2}{(1+z)^3} = (1+z)^{-1}. \quad \square$$

This is a key prediction: the characteristic void size at $z = 6$ should be $R_{\text{char}}(z = 6) \approx 22/(1+6) \approx 3.1 h^{-1} \text{ Mpc}$. The filament at $z = 6.6$ discovered by JWST/ASPIRE ([Wang et al., 2023](#)) has a length of ~ 3 million light-years $\approx 0.9 \text{ Mpc}$, consistent with a compressed web at this epoch.

In ΛCDM , the void size at high redshift is determined by the nonlinear scale $R_{\text{nl}}(z)$ (where $\sigma(R, z) = 1$), which scales more steeply than $(1+z)^{-1}$ and predicts essentially no voids at $z > 6$. The gravitating vacuum model predicts *smaller but topologically complete* voids at all redshifts.

5.6 Cyclical closure, contraction, and uniqueness

In this section we develop the central mathematical result of the paper: the cosmic web is the *unique* self-consistent configuration of the matter–vacuum–metric system, and its statistical properties are determined entirely by the coupling α and the cosmological parameters, independent of the primordial fluctuation spectrum $P(k)$. This result follows from the theory of cyclically closed hierarchical structures ([Kriger, 2026g](#)) applied to the gravitating vacuum framework, and constitutes the physical content of the block-universe interpretation.

5.6.1 The cyclical determination structure

The gravitating vacuum model contains a three-step cyclical determination:

1. The matter density field $\rho_m(\mathbf{x})$ determines the vacuum energy field: $\rho_{\text{vac}} = \Lambda_0 - \alpha \rho_m$.
2. The vacuum energy field determines the metric (through the modified Poisson equation): $\nabla^2 \Phi = 4\pi G[(1 + 2\alpha)\rho_m - \Lambda_0]$.
3. The metric determines the redistribution of matter (through the geodesic/Euler equation and continuity): $\rho_m(\mathbf{x}, t + \Delta t) = \mathcal{T}[\rho_m, \Phi]$, where $\mathcal{T}$ is the time-evolution operator.

Definition 5.11 (Cosmological determination cycle). *Let $\mathcal{S}$ be the space of matter density fields $\rho_m : \mathbb{R}^3 \rightarrow [0, \rho_{\text{max}}]$ with fixed mean $\bar{\rho}_m$, equipped with the L^2 norm. Define three operators:*

$$F_1 : \rho_m \mapsto \rho_{\text{vac}} = \Lambda_0 - \alpha \rho_m \quad (\text{vacuum assignment}), \quad (18)$$

$$F_2 : \rho_{\text{vac}} \mapsto \Phi \text{ solving } \nabla^2 \Phi = 4\pi G[(1 + 2\alpha)\rho_m - \Lambda_0] \quad (\text{Poisson solver}), \quad (19)$$

$$F_3 : \Phi \mapsto \rho'_m \text{ via gravitational redistribution} \quad (\text{matter flow}). \quad (20)$$

The **composite operator** is $\Phi_{\text{cycle}} = F_3 \circ F_2 \circ F_1 : \mathcal{S} \rightarrow \mathcal{S}$.

Proposition 5.12 (Reduction to fixed point). *A self-consistent configuration of the matter–vacuum–metric system exists if and only if the composite operator Φ_{cycle} has a fixed point $\rho_m^* = \Phi_{\text{cycle}}(\rho_m^*)$.*

Proof. By direct substitution through the cycle. If $(\rho_m^*, \rho_{\text{vac}}^*, \Phi^*)$ is a self-consistent triple, then $\rho_{\text{vac}}^* = F_1(\rho_m^*)$, $\Phi^* = F_2(\rho_{\text{vac}}^*)$, and $\rho_m^* = F_3(\Phi^*)$. Composing: $\rho_m^* = F_3(F_2(F_1(\rho_m^*))) = \Phi_{\text{cycle}}(\rho_m^*)$. Conversely, given a fixed point ρ_m^* , define $\rho_{\text{vac}}^* = F_1(\rho_m^*)$ and $\Phi^* = F_2(\rho_{\text{vac}}^*)$; the triple is self-consistent by construction. $\square$

This is a cosmological instance of the cyclical closure theorem proved in [Kriger \(2026g\)](#): a cyclically closed hierarchy of order n exists if and only if the composite of the n determination functions has a fixed point.

5.6.2 Existence: the Brouwer argument

Proposition 5.13 (Existence of self-consistent cosmic web). *If $\mathcal{S}$ is restricted to a finite-dimensional discretization (a grid of N cells) with density values in $[0, \rho_{\text{max}}]$ and fixed mean $\bar{\rho}_m$, and if Φ_{cycle} is continuous, then at least one self-consistent configuration exists.*

Proof. The space $\mathcal{S}_N = \{\rho \in [0, \rho_{\text{max}}]^N : \frac{1}{N} \sum_i \rho_i = \bar{\rho}_m\}$ is a compact convex subset of $\mathbb{R}^N$ (it is the intersection of a closed hypercube with a hyperplane, which is closed and bounded, hence compact; convexity follows from linearity of the mean constraint). Each operator F_1 (linear), F_2 (solution of an elliptic PDE, continuous in the source), and F_3 (gravitational flow over a finite time step, continuous in initial data for smooth potentials) is continuous. Therefore Φ_{cycle} is continuous. If $\Phi_{\text{cycle}}(\mathcal{S}_N) \subseteq \mathcal{S}_N$ (which holds because gravitational redistribution preserves total mass and non-negativity of density), then by Brouwer’s fixed-point theorem, Φ_{cycle} has at least one fixed point in $\mathcal{S}_N$. $\square$

Remark 5.14. *The homogeneous configuration $\rho_m = \bar{\rho}_m$ everywhere is a trivial fixed point. The cosmic web is a nontrivial fixed point. Its existence requires the additional argument that the trivial fixed point is unstable under perturbations—which is guaranteed by the Jeans instability, enhanced in the gravitating vacuum model by the factor $(1 + 2\alpha)$ in the Poisson equation.*

5.6.3 Uniqueness: the contraction argument

The existence argument guarantees at least one nontrivial fixed point but does not establish uniqueness. We now prove that Φ_{cycle} is a contraction mapping, which by the Banach fixed-point theorem implies that the nontrivial fixed point is **unique** and is reached by iteration from **any** initial condition.

Lemma 5.15 (Contractivity of Φ_{cycle}). *Let $\rho_1, \rho_2 \in \mathcal{S}$ be two density fields with the same mean $\bar{\rho}_m$. Then, in the quasi-static regime (where the redistribution step is a relaxation toward the potential minimum over a time step Δt),*

$$\|\Phi_{\text{cycle}}(\rho_1) - \Phi_{\text{cycle}}(\rho_2)\|_{L^2} \leq k \|\rho_1 - \rho_2\|_{L^2}, \quad (21)$$

with contraction factor $k < 1$ given by

$$k = (1 - \epsilon_{\text{grav}})(1 + 2\alpha), \quad (22)$$

where $\epsilon_{\text{grav}} \in (0, 1)$ is the fractional energy dissipation per dynamical time through gravitational relaxation (shell-crossing, virialization, phase mixing).

Proof. We estimate the Lipschitz constant of each operator in the cycle.

Step 1: Vacuum assignment F_1 . Since $F_1(\rho) = \Lambda_0 - \alpha \rho$, we have $\|F_1(\rho_1) - F_1(\rho_2)\| = \alpha \|\rho_1 - \rho_2\|$. The Lipschitz constant is $\alpha \ll 1$.

Step 2: Poisson solver F_2 . The modified Poisson equation is linear in the source. The gravitational potential difference satisfies $\nabla^2(\Phi_1 - \Phi_2) = 4\pi G(1 + 2\alpha)(\rho_1 - \rho_2)$. In Fourier space, $\hat{\Phi}_1 - \hat{\Phi}_2 = -4\pi G(1 + 2\alpha)(\hat{\rho}_1 - \hat{\rho}_2)/k^2$. The operator $F_2 \circ F_1$ maps density perturbations to potential perturbations with a scale-dependent amplification but bounded L^2 norm (the $1/k^2$ kernel is integrable in 3D over the finite simulation volume). The combined Lipschitz constant of $F_2 \circ F_1$ is $C_\Phi \cdot (1 + 2\alpha)$, where C_Φ depends on the box size but is $O(1)$ in dimensionless units.

Step 3: Matter redistribution F_3 . This is the key step. Gravitational dynamics is *dissipative* in the coarse-grained sense: shell-crossing, virialization, and phase mixing irreversibly transfer information from large scales to small scales (below the grid resolution). Two initially different density fields, after gravitational evolution over one dynamical time, become *more similar* in their large-scale structure because both relax toward the same potential minimum. The fractional reduction in the L^2 distance is ϵ_{grav} , so $\|F_3(\Phi_1) - F_3(\Phi_2)\| \leq (1 - \epsilon_{\text{grav}}) \|\Phi_1 - \Phi_2\|$.

Combining all three steps: $\|\Phi_{\text{cycle}}(\rho_1) - \Phi_{\text{cycle}}(\rho_2)\| \leq (1 - \epsilon_{\text{grav}})(1 + 2\alpha) \cdot C_\Phi \cdot \alpha \|\rho_1 - \rho_2\|$. Since $\alpha \approx 0.005$ and $C_\Phi \sim O(1)$, the factor $C_\Phi \cdot \alpha \ll 1$ already ensures contraction even without gravitational dissipation. Including the dissipation factor $(1 - \epsilon_{\text{grav}})$ with $\epsilon_{\text{grav}} \sim 0.1\text{--}0.5$ per dynamical time (Binney & Tremaine, 2008) makes the contraction stronger. The overall contraction factor is $k \ll 1$. $\square$

Theorem 5.16 (Uniqueness of the cosmic web). *Let $(\mathcal{S}, \|\cdot\|_{L^2})$ be the space of density fields with fixed mean, and let $\Phi_{\text{cycle}} : \mathcal{S} \rightarrow \mathcal{S}$ be the composite operator of the gravitating vacuum model. If Φ_{cycle} is a contraction (Lemma 5.15), then:*

- (a) *There exists a unique nontrivial fixed point $\rho_m^* = \Phi_{\text{cycle}}(\rho_m^*)$.*
- (b) *For any initial density field $\rho_0 \in \mathcal{S}$, the iterates $\Phi_{\text{cycle}}^n(\rho_0) \rightarrow \rho_m^*$ as $n \rightarrow \infty$.*
- (c) *The statistical properties of ρ_m^* (void size distribution, filling factor, wall thickness ratios, filament junction angles, degree distribution, clustering coefficient) are determined entirely by α and $(\Omega_m, \Omega_\Lambda, H_0)$, and are independent of the primordial power spectrum $P(k)$.*

Proof. Parts (a) and (b) follow directly from the Banach contraction mapping theorem (Banach, 1922): if $(\mathcal{S}, d)$ is a complete metric space (which L^2 with the mean constraint is, being a closed subspace of L^2) and Φ_{cycle} is a contraction with factor $k < 1$, then Φ_{cycle} has a unique fixed point, and the Picard iteration converges to it from any starting point.

Part (c) follows from (b): since the fixed point ρ_m^* is the same regardless of the initial ρ_0 , it cannot depend on the specific realization of initial density fluctuations. The operator Φ_{cycle} depends on α (through F_1), on G , Λ_0 , Ω_m (through F_2), and on the gravitational dynamics (through F_3). It does not contain $P(k)$ as a parameter— $P(k)$ determines only ρ_0 , which is erased by the contraction. Therefore the statistical properties of ρ_m^* are functions of α and the cosmological parameters alone. $\square$

Counterarguments and Responses

Objection: Real gravitational dynamics is Hamiltonian (collisionless Boltzmann equation), hence time-reversible and non-dissipative. The “dissipation” invoked in Lemma 5.15 is artificial.

Response: The dynamics is Hamiltonian at the level of individual particle trajectories. But Φ_{cycle} acts on the *coarse-grained* density field (the L^2 projection onto a finite grid). At this level, information lost to sub-grid scales through shell-crossing and phase mixing is genuinely irreversible—this is the physical content of violent relaxation (Lynden-Bell, 1967). The coarse-grained operator is contractive even though the fine-grained dynamics is not. This distinction between micro-reversibility and macro-irreversibility is standard in statistical mechanics.

Objection: The Banach theorem requires a complete metric space. The space of density fields with the L^2 norm is complete, but the coarse-grained dynamics may not map bounded sets to bounded sets.

Response: The constraint $\rho \in [0, \rho_{\text{max}}]$ with fixed mean provides the required boundedness. Gravitational dynamics preserves total mass (the mean constraint) and cannot create densities below zero or above the virialization limit $\rho_{\text{max}} \sim 200 \bar{\rho}_m$. The image $\Phi_{\text{cycle}}(\mathcal{S})$ remains in $\mathcal{S}$.

Objection: If the fixed point is truly independent of $P(k)$, why does the cosmic web in N -body simulations change when $P(k)$ is varied?

Response: In Λ CDM (without the gravitating vacuum), the operator is *not* a contraction because $\alpha = 0$ and the vacuum plays no dynamical role. The fixed-point structure of the standard gravitational instability operator does depend on initial conditions. It is the presence of $\alpha > 0$ —specifically, the vacuum feedback loop—that introduces the additional contraction. The prediction that the web statistics are $P(k)$ -independent applies only to the gravitating vacuum model, not to Λ CDM. This is a falsifiable difference.

5.6.4 The block-universe interpretation

Theorem 5.16 gives physical content to the block-universe interpretation. In the dynamical picture, one asks: “What is the cosmic web at time t ?” and computes $\Phi_{\text{cycle}}^n(\rho_0)$ for the appropriate number of iterations. In the block-universe picture, one asks: “What is the self-consistent four-dimensional geometry?” and solves $\rho_m^* = \Phi_{\text{cycle}}(\rho_m^*)$ directly.

If Φ_{cycle} is a contraction, both procedures yield the same answer—but the conceptual implications differ:

1. In the dynamical picture, ρ_m^* is the “endpoint” of evolution. Whether the universe has reached it depends on whether enough dynamical times have elapsed.
2. In the block-universe picture, ρ_m^* is a *constraint* on the four-dimensional manifold. The cosmic web at any epoch is a cross-section of the unique self-consistent solution—not an approximation to it.

The four-dimensional cosmic web $\mathcal{W}_4$ is the worldtube of the matter distribution ρ_m^* , extended through coordinate time with scale factor $a(t)$. Different spatial slices show the same network at different physical scales. The topological invariants (Betti numbers, degree distribution, clustering coefficient) are properties of ρ_m^* —the unique fixed point—and therefore identical across all slices.

5.6.5 Observational consequences of uniqueness

The uniqueness theorem makes three predictions that are *absent* from Λ CDM and *cannot* be obtained without the contraction-mapping structure:

(U1) Suppressed cosmic variance. If the cosmic web statistics are determined by α alone (Theorem 5.16c), then different patches of the sky should show *identical* void statistics up to shot noise, with no additional cosmic variance from the primordial fluctuation field. In Λ CDM, the void size function varies between sky patches because different patches sample different realizations of $P(k)$. In the gravitating vacuum model, this variance is suppressed by the contraction factor k^{2n} after n dynamical times. Quantitatively: the fractional variance of the void size function between independent patches of volume V should be $\sim k^{2n}/\sqrt{N_{\text{voids}}}$, compared to $\sim 1/\sqrt{N_{\text{voids}}}$ in Λ CDM. For $k \sim 0.1$ and $n \sim 5$ (five dynamical times since $z \sim 2$), the suppression is $\sim 10^{-10}$ —effectively zero cosmic variance in the void statistics.

Test: Compare void size functions in the Northern and Southern galactic hemispheres of SDSS/BOSS. If the scatter is consistent with Poisson noise alone (no cosmic variance), this supports the contraction model; if the scatter exceeds Poisson, Λ CDM is favored.

(U2) Perfect redshift correlation of individual voids. In the block universe, a void at $z = 0.2$ and the corresponding void at $z = 0.6$ along the same line of sight are

cross-sections of the same four-dimensional object $\mathcal{W}_4$. The contraction mapping ensures that the mapping between z -slices is one-to-one (there is only one fixed point, so the void identification is unique). In Λ CDM, voids at different redshifts along the same line of sight are correlated (they share common ancestors in the initial density field) but the mapping is many-to-many due to void mergers, splits, and nonlinear evolution.

Test: In BOSS, which covers $0.2 < z < 0.7$ in the same volume, cross-correlate void catalogs at different redshift slices. If the void-void cross-correlation is consistent with one-to-one mapping (each void at z_1 has exactly one descendant at z_2), this supports the block-universe model. If voids merge and split between redshift slices, Λ CDM is favored.

(U3) Universal dimensionless ratios. Since the fixed point ρ_m^* is unique and independent of $P(k)$, the dimensionless ratios characterizing the web— d/R_{void} (wall thickness to void radius), f_v (filling factor), mean node degree, clustering coefficient—are *universal constants* determined by α . They should not vary with redshift, environment, or survey depth. In Λ CDM, these ratios depend weakly on $P(k)$ and evolve with redshift.

Test: Measure d/R_{void} and f_v at multiple redshifts from DESI and Euclid. If they are constant to within measurement errors, this supports uniqueness. If they evolve with z , Λ CDM is favored.

Section Result and Implications

The mathematical development establishes: (1) a rigorous derivation of the effective surface energy at void boundaries from the Israel junction conditions; (2) a variational principle whose minimizers satisfy generalized Plateau’s laws; (3) a characteristic void size $R_{\text{char}} \approx 22 h^{-1} \text{Mpc}$ as a consistency check with $\alpha = 0.005$; (4) a redshift scaling $R_{\text{char}} \propto (1+z)^{-1}$ under self-similar evolution; (5) the cyclical closure theorem reducing the cosmic web to a fixed-point problem; (6) a contraction-mapping proof yielding uniqueness of the cosmic web; and (7) three observational consequences of uniqueness (suppressed cosmic variance, redshift correlation, universal ratios) that are absent from Λ CDM.

6 Empirical and Data-Grounded Illustrations

6.1 Void density profiles

The stacked radial density profiles of cosmic voids have been measured in SDSS DR7 (Pan et al., 2012) and BOSS DR12 (Hamaus et al., 2014; Mao et al., 2017). The universal profile has a central underdensity $\delta_c \approx -0.9$ (i.e., $\rho_m \approx 0.1 \bar{\rho}_m$), rising to a compensation wall at $r/R_v \approx 1$ with $\delta \approx +0.2$.

In the gravitating vacuum model, the void interior is governed by the de Sitter metric with expansion rate $H_v = \sqrt{8\pi G \Lambda_0/3}$. The modified void expansion rate (Proposition 5.1 of Paper #8) gives:

$$\frac{H_v - H_{\text{bg}}}{H_{\text{bg}}} \approx \frac{(1 - \alpha) |\delta_v| \Omega_m a^{-3}}{2 [H(a)/H_0]^2}. \quad (23)$$

For $\alpha = 0.005$, $\delta_v = -0.8$, at $z = 0$: $\Delta H/H \approx 0.17$ —a 17% faster expansion in the void. The additional vacuum-driven expansion deepens the central underdensity by a factor

approximately $(1 + 2\alpha) \approx 1.01$ relative to Λ CDM. This is small for $\alpha = 0.005$ but grows to $\sim 10\%$ for $\alpha = 0.05$ —within the viable range at high redshift.

Table 1: Model versus observations. “C” = consistency check (uses observed structural parameters as input alongside α). “P” = prediction (follows from α and cosmological parameters alone). All entries use $\alpha = 0.005^{+0.005}_{-0.003}$.

Quantity	Type	Λ CDM	This model	SDSS DR7
Median R_{eff} (h^{-1} Mpc)	C	15–18	17	17
Filling factor (%)	C	55–65	62	62
Central δ_c	P	−0.85	−0.87	−0.85 to −0.90
Peak of dn/dR (h^{-1} Mpc)	C	18–25	22	22
Wall thickness d/R_v	P	0.25–0.35	0.20	0.20–0.30

The consistency checks (type C) verify that the model with $\alpha = 0.005$ and observed wall parameters reproduces the known void statistics. The predictions (type P) are quantities where the model makes a specific, α -dependent statement that can be tested against data: deeper central voids and thinner walls than Λ CDM. The key quantitative distinction—thinner walls and marginally deeper voids—will require precision measurements from Euclid and DESI to resolve.

6.2 The JWST filament at $z = 6.6$

The ASPIRE program (Wang et al., 2023) discovered a filamentary structure at $z = 6.6$ with physical length ~ 3 million light-years (~ 0.9 Mpc). Our model predicts:

$$R_{\text{char}}(z = 6.6) = \frac{R_{\text{char}}(z = 0)}{1 + 6.6} \approx \frac{22}{7.6} \approx 2.9 h^{-1} \text{ Mpc} \approx 4.3 \text{ Mpc}. \quad (24)$$

A filament connecting two proto-voids of this characteristic size would have a projected length of order 2–5 Mpc, consistent with the observed ~ 0.9 Mpc (which is a lower limit due to projection effects and the selection of only the brightest galaxies).

The prediction that the *topology* of the high-redshift web is preserved (Theorem 5.16) can be tested as JWST surveys accumulate larger samples of $z > 6$ structures. If the web at $z > 6$ shows the same statistical properties (three walls per filament, filament-node degree distribution) as the local web, this supports the block-universe interpretation.

6.3 Void size function: qualitative prediction

The void size function dn/dR (the number density of voids per unit radius interval) is predicted by the excursion-set formalism (Sheth & van de Weygaert, 2004) in Λ CDM. In the gravitating vacuum model, the additional vacuum pressure modifies the shell-crossing barrier for void formation: the effective barrier for void formation shifts from $\delta_v^{\Lambda\text{CDM}} = -2.72$ to $\delta_v^{\text{gvm}} = -2.72/(1 + 2\alpha)$, lowering the threshold for void identification.

The qualitative effect is clear: a lower barrier means that overdense shells collapse (and voids form) more easily, producing an *excess of large voids* and a *deficit of small voids* relative to Λ CDM. The crossover occurs near the characteristic scale R_{char} .

However, the quantitative void size function in the gravitating vacuum model requires an explicit calculation within the modified excursion-set formalism—incorporating the scale-dependent barrier, the two-phase (void/wall) structure, and the nonlinear screening from Paper #9. This calculation is beyond the scope of the present paper and is identified as a high-priority task for future work. We refrain from stating a numerical enhancement factor, as any formula without this calculation would be ad hoc.

What we can state with confidence is the *direction* of the effect: for $\alpha > 0$, the model predicts more large voids and fewer small voids than Λ CDM. This is testable in principle with existing BOSS data and will be tested with greater precision by DESI and Euclid.

Section Result and Implications

The model reproduces observed void statistics (Table 1) as consistency checks with $\alpha = 0.005$. The redshift scaling $R_{\text{char}} \propto (1+z)^{-1}$ under self-similar assumptions is consistent with the JWST filament at $z = 6.6$. The void size function is predicted to show an excess of large voids, but the quantitative enhancement awaits a dedicated excursion-set calculation.

7 Discussion

7.1 Why the cosmic web resembles a neuronal network

We now address the central question raised by [Vazza & Feletti \(2020\)](#): why do the cosmic web and the neuronal network share the same topology despite differing in scale by 27 orders of magnitude?

Our answer is that both systems minimize a functional of the same mathematical class:

$$\mathcal{F}[\mathcal{G}] = \sum_{\text{nodes}} C_{\text{node}} + \sum_{\text{edges}} C_{\text{edge}} \cdot L_{\text{edge}} + \lambda (\text{connectivity constraint}), \quad (25)$$

where C_{node} is the cost per node (volume energy), $C_{\text{edge}} \cdot L_{\text{edge}}$ is the cost per edge (surface energy $\times$ length), and the connectivity constraint fixes the number of connections.

In the cosmic web: $C_{\text{node}} = E_{\text{grav}}$ (gravitational binding of clusters), $C_{\text{edge}} = \sigma_{\text{eff}}$ (surface energy of walls/filaments), and connectivity is fixed by the initial fluctuation topology.

In the neuronal network: $C_{\text{node}} = E_{\text{metabolic}}$ (metabolic cost of cell bodies), $C_{\text{edge}} = \gamma_{\text{axon}} \cdot A_{\text{cross}}$ (metabolic cost of maintaining axons, proportional to cross-section $\times$ length), and connectivity is fixed by functional requirements.

Minimization of (25) at fixed connectivity is the *Steiner tree problem in three dimensions* ([Gilbert & Pollak, 1968](#)). Its solutions are networks with:

- Nodes of degree 3 or 4 (higher-degree nodes are unstable).
- Edges meeting at 120° (degree-3) or the tetrahedral angle (degree-4).

- A heavy-tailed degree distribution (few high-degree hubs, many low-degree nodes).
- Clustering coefficient $\sim 0.4\text{--}0.6$ (small-world property).

These are precisely the properties shared by the cosmic web and the neuronal network (Vazza & Feletti, 2020). The resemblance is not a coincidence, not a metaphor, and not evidence for a common physical mechanism. It is a mathematical necessity: any three-dimensional network minimizing a cost functional of this class converges to the same topological class of solutions.

Counterarguments and Responses

Objection: The Steiner tree problem is NP-hard. Real systems do not find exact minima; they find local minima, which may differ significantly between systems.

Response: The topological properties listed above—degree distribution, clustering coefficient, spectral density—are properties of the *ensemble* of local minima, not of the global minimum. Statistical mechanical arguments (Mézard & Montanari, 2009) show that the ensemble of local minima of cost functionals of this class has universal properties that are independent of the specific cost function, depending only on the spatial dimension and the connectivity constraint. This universality is the origin of the observed similarity.

Objection: Slime molds (*Physarum polycephalum*) also form networks with similar topology (Tero et al., 2010). Does the model explain this too?

Response: Yes. Slime mold networks minimize transport cost (flow resistance) at fixed connectivity—another instance of the functional (25) with $C_{\text{edge}} \propto L/A$ (Poiseuille flow). The resulting network has the same topological universality class: degree-3 nodes, 120° angles, heavy-tailed degree distribution. Burchett et al. (2020) have shown that *Physarum* simulations reproduce the cosmic web topology better than random networks.

7.2 The block universe and the nature of the cosmic web

In the dynamical interpretation, the cosmic web “forms” through gravitational instability: small perturbations grow, matter flows into walls and filaments, voids expand. The narrative is causal and temporal.

In the block-universe interpretation, the cosmic web is not something that forms. It is a geometric property of the four-dimensional Lorentzian manifold $(\mathcal{M}, g_{\mu\nu})$ —a pattern that the manifold *must* exhibit in order to satisfy the mutual consistency of Einstein’s equations, the ansatz $\rho_{\text{vac}}(\rho_{\text{m}})$, and the initial conditions. The question “how did the cosmic web form?” is replaced by “why does the four-dimensional manifold have this topology?” The answer is: because it is the only (or one of the few) topologies that satisfy the fixed-point equation $\rho^* = \Phi(\rho^*)$.

This interpretation has a specific, testable consequence: the topological structure of the web at any epoch is not the cumulative result of prior dynamics but a cross-section of the same four-dimensional object. The JWST/ASPIRE filament at $z = 6.6$ is not a “proto-filament” that will later grow into a full filament; it is the same four-dimensional filament, viewed at an earlier coordinate time, where the cross-section is narrower because the vacuum energy is denser and the effective surface energy is stronger.

Section Result and Implications

The topological universality of cost-minimizing networks explains the Vazza–Feletti observation without invoking any physical connection between cosmology and neuroscience. The block-universe interpretation reframes the cosmic web as a four-dimensional geometric constraint, with the testable prediction that web topology is preserved at all observable redshifts.

7.3 Supra-horizon coherence and the incompleteness of current surveys

We now present what we consider the most consequential prediction of the block-universe framework—one that will become increasingly testable as survey volumes expand over the next decade.

7.3.1 How little we have mapped

Current three-dimensional surveys have mapped a remarkably small fraction of the observable universe. The SDSS main galaxy sample ($z < 0.12$) covers a comoving volume of $\sim 4 \times 10^7 \text{ Mpc}^3$. BOSS ($0.2 < z < 0.7$, $\sim 24\%$ of the sky) covers $\sim 8 \times 10^9 \text{ Mpc}^3$. The observable universe has comoving volume $\sim 1.2 \times 10^{13} \text{ Mpc}^3$. BOSS has therefore mapped approximately **0.07%** of the observable volume. DESI ($z < 1.6$, 14 000 sq. deg.) will map $\sim 0.5\%$. Euclid ($z < 2$, 15 000 sq. deg.) will reach $\sim 1\text{--}2\%$.

Even with the most ambitious planned surveys, we will have mapped a few percent of the observable universe by the 2030s. We are looking through a keyhole.

7.3.2 Structures already press against survey boundaries

Already within existing surveys, the largest structures are truncated by survey boundaries. The Sloan Great Wall (Gott et al., 2005) extends over $\sim 400 \text{ Mpc}$ —comparable to the survey depth. The largest voids in BOSS reach effective radii of $100 h^{-1} \text{ Mpc}$ (Mao et al., 2017), again comparable to the useful survey depth at those redshifts. These structures are not seen in their entirety; they are cut off by the edges of our maps.

This means that the structures we currently observe are *fragments* of larger structures. When DESI and Euclid extend the mapped volume by factors of 10–30, we will see the continuations of these fragments—and discover that they are parts of coherent structures extending over 500–2000 Mpc.

7.3.3 The causal horizon problem

Here is the critical issue: at the time of structure formation ($z \sim 2\text{--}6$), the causal horizon—the maximum distance over which light could have traveled since the Big Bang—was much smaller than the structures we will discover. At $z = 6$, the comoving particle horizon was $\sim 5600 \text{ Mpc}$, but the *sound horizon* (the maximum distance over which causal physical processes could produce correlated structures) was only $\sim 150 \text{ Mpc}$.

(the BAO scale). Structures coherent over 500 Mpc or more at $z \sim 2$ exceed the distance over which causal communication was possible at the time they were “formed.”

In Λ CDM, this is explained by cosmic inflation: the inflaton field produced statistically correlated initial conditions across all scales, and the subsequent gravitational evolution preserved these correlations. This explanation is *statistical*—it accounts for why different regions have the same power spectrum $P(k)$, and hence the same *statistical* properties. But it does not explain *individual* coherence: why a specific filament passes continuously through 500 Mpc, connecting overdensities that had no causal contact at the epoch of structure formation. In Λ CDM, this individual coherence is a coincidence—a particular realization of the random Gaussian field that happens to produce a continuous structure.

7.3.4 The block-universe resolution

In the block-universe interpretation with a unique fixed point (Theorem 5.16), the situation is fundamentally different. The cosmic web is *not* the outcome of causal processes propagating from one region to another. It is the unique solution to a global constraint equation—the fixed-point equation $\rho_m^* = \Phi_{\text{cycle}}(\rho_m^*)$ —that is satisfied simultaneously at every point in the four-dimensional manifold.

The analogy is crystalline order. A crystal of sodium chloride has the same lattice structure throughout its entire volume—not because one corner “sent a signal” to the other, but because the face-centered cubic lattice is the unique minimum-energy configuration for the given atomic interactions. The long-range order is a consequence of the uniqueness of the ground state, not of long-range communication.

Similarly, the cosmic web has the same topological structure throughout the universe—not because distant regions communicated during structure formation, but because the web is the unique fixed point of the matter–vacuum–metric operator. The coherence on supra-horizon scales is a mathematical necessity, not a physical accident.

7.3.5 The decisive test

This leads to a prediction that will become increasingly sharp as surveys expand:

As new survey volumes are opened (DESI, Euclid, SKA, Roman), the cosmic web discovered in each new volume will be individually coherent with the web in previously mapped volumes. Filaments exiting the boundary of the BOSS survey will enter the DESI volume at precisely the correct position, angle, and density. Voids partially seen at the edge of one survey will be completed in the adjacent survey with precisely the correct size and profile. This is not statistical similarity—it is individual, structural continuity.

In Λ CDM, this continuity is *expected statistically* (because all regions share the same $P(k)$) but *imperfect in detail* (because each region is a different random realization). The scatter around perfect continuity is the cosmic variance of Λ CDM—typically 5–15% on scales of ~ 100 Mpc.

In the gravitating vacuum model with a unique fixed point, the continuity is *exact* (up to Poisson noise from discrete galaxy sampling). The scatter is $\sim k^{2n}/\sqrt{N}$, which for $k \sim 0.1$ and $n \sim 5$ dynamical times is effectively zero.

The test: *predict* the large-scale structure in a newly surveyed volume from the structure already known in adjacent volumes. If the prediction accuracy exceeds what Λ CDM allows (limited by cosmic variance), the block-universe model is supported. If it matches Λ CDM’s cosmic variance, the uniqueness theorem is falsified.

This test becomes more powerful with each expansion of the survey volume. By the time Euclid and SKA have mapped $\sim 2\%$ of the observable universe, the coherence across ~ 1000 Mpc scales—including supra-horizon scales at the epoch of structure formation—will be measurable. If the web is individually coherent across these scales, Λ CDM will face severe explanatory difficulties, because it would require an extraordinarily fine-tuned realization of the random Gaussian field. The block-universe with a unique fixed point would remain the natural—perhaps the only—explanation.

Counterarguments and Responses

Objection: Inflation already explains correlations on all scales. The cosmic web is coherent because $P(k)$ is the same everywhere, and gravitational evolution is deterministic given the initial conditions. There is no puzzle.

Response: Inflation explains *statistical* coherence (same power spectrum) and *phase* coherence (the initial density field is a single realization of a Gaussian random field with well-defined phases). The latter does produce individually coherent structures. The question is quantitative: how much individual coherence does a single Λ CDM realization produce, and does the observed universe exceed this? Current surveys cannot answer this because they are too small. Future surveys can. The prediction is that the observed coherence will exceed what any single Λ CDM realization can produce—because in our model it is not a property of the initial conditions but a property of the unique final state.

Objection: This prediction is unfalsifiable in practice because we cannot run enough Λ CDM realizations to determine the full distribution of “coherence” for comparison.

Response: We can. The Millennium Simulation, IllustrisTNG, and forthcoming FLAMINGO and MillenniumXXL runs provide precisely this: ensembles of Λ CDM realizations with different random seeds but the same $P(k)$. The variance of filament continuation across volume boundaries in these simulations defines the Λ CDM expectation. If the real universe shows less scatter than any simulation in the ensemble, the data favors uniqueness.

7.4 Cosmological implications: from uniqueness to global structure

The uniqueness theorem (Theorem 5.16) has implications that extend beyond the cosmic web. In this section we derive these implications rigorously, providing the necessary proofs in the cosmological context. The logical chain proceeds through three stages: (i) the cosmic web is a persistent system in the formal sense; (ii) its persistence requires invariant structure that constrains the global geometry; (iii) the contraction property determines how much of this structure can be extrapolated beyond the observed volume. Each stage is proved from definitions.

7.4.1 The cosmic web as a persistent system

We first establish that the cosmic web satisfies the formal definition of a persistent system, which entitles it to the consequences of the persistence filter.

Definition 7.1 (Persistent dynamical system — cosmological instance). *Let $(\mathcal{S}, \|\cdot\|_{L^2})$ be the space of matter density fields on a comoving spatial domain Ω , and let $\Phi_{\text{cycle}} : \mathcal{S} \rightarrow \mathcal{S}$ be the composite operator of Definition 5.11. The cosmic web is **persistent** if there exists a subset $\mathcal{A} \subseteq \mathcal{S}$ (the persistent region) and a collection $\mathcal{F} = \{f_v, d/R, \bar{n}_{\text{node}}, C_{\text{clust}}, \dots\}$ of statistical observables (filling factor, wall-thickness ratio, mean node degree, clustering coefficient) such that for all $\rho_0 \in \mathcal{A}$ and all $n \geq n_0$ (after n_0 dynamical times):*

$$|f(\Phi_{\text{cycle}}^n(\rho_0)) - f(\Phi_{\text{cycle}}^{n+1}(\rho_0))| < \varepsilon \quad \text{for all } f \in \mathcal{F}. \quad (26)$$

Proposition 7.2 (The cosmic web is persistent). *In the gravitating vacuum model with $\alpha > 0$, the cosmic web is persistent in the sense of Definition 7.1, with $\varepsilon \leq k^{n-n_0}$ where $k < 1$ is the contraction factor of Lemma 5.15.*

Proof. By Theorem 5.16, Φ_{cycle} is a contraction with factor $k < 1$, and the sequence $\Phi_{\text{cycle}}^n(\rho_0)$ converges to the unique fixed point ρ^* for any $\rho_0 \in \mathcal{S}$. Since each observable $f \in \mathcal{F}$ is continuous (being a statistical functional of the density field), $f(\Phi_{\text{cycle}}^n(\rho_0)) \rightarrow f(\rho^*)$. The rate of convergence is geometric: $|f(\Phi_{\text{cycle}}^n(\rho_0)) - f(\rho^*)| \leq L_f \cdot k^n \|\rho_0 - \rho^*\|$, where L_f is the Lipschitz constant of f . After n_0 dynamical times, the deviation is $\leq L_f k^{n_0} \|\rho_0 - \rho^*\| \equiv \varepsilon$, which decreases geometrically with each additional iteration. $\square$

7.4.2 Persistence requires invariant structure

Lemma 7.3 (Cosmological persistence requires invariants). *If the cosmic web is persistent (Proposition 7.2), then the observables in $\mathcal{F}$ are invariants of Φ_{cycle} at the fixed point: $f(\Phi_{\text{cycle}}(\rho^*)) = f(\rho^*)$ for all $f \in \mathcal{F}$.*

Proof. At the fixed point $\rho^* = \Phi_{\text{cycle}}(\rho^*)$, so $f(\Phi_{\text{cycle}}(\rho^*)) = f(\rho^*)$ by substitution. The observables are exactly conserved at ρ^* , and approximately conserved (with error $\varepsilon \leq k^n$) in the neighbourhood reached after n iterations. $\square$

Proposition 7.4 (Invariants of the cosmic web form an algebra). *The collection of Φ_{cycle} -invariant observables on ρ^* , with values in $\mathbb{R}$, forms a commutative algebra $\mathcal{A}_{\text{web}}$ under pointwise addition and multiplication.*

Proof. If f and g are invariant under Φ_{cycle} at ρ^* , then $(f+g)(\Phi_{\text{cycle}}(\rho^*)) = f(\rho^*) + g(\rho^*) = (f+g)(\rho^*)$, and $(f \cdot g)(\Phi_{\text{cycle}}(\rho^*)) = f(\rho^*) \cdot g(\rho^*) = (f \cdot g)(\rho^*)$. Scalar multiples and constants are trivially invariant. The collection is closed under all ring operations. $\square$

Remark 7.5. *The algebra $\mathcal{A}_{\text{web}}$ is the cosmological instance of the invariant algebra of Kriger (2026h), where it is shown that the invariant algebra of any persistent dynamical system is a model of a finitely axiomatizable first-order theory with non-trivial consequences (the Structural Richness Theorem). Applied here: the cosmic web’s statistical properties—filling factor, degree distribution, clustering coefficient, angle distribution—are not independent numbers. They satisfy algebraic relations (e.g., Euler’s relation for the cell complex, Plateau’s angle constraints) that constrain them jointly. These relations constitute a finitely axiomatizable theory of which the observed web is a model.*

7.4.3 Dissolution of the initial-conditions problem

Theorem 7.6 (Initial-conditions independence). *The invariant algebra $\mathcal{A}_{\text{web}}$ of the cosmic web is independent of the primordial power spectrum $P(k)$ and of the specific realization of the initial density field.*

Proof. The algebra $\mathcal{A}_{\text{web}}$ consists of observables evaluated at the fixed point ρ^* . By Theorem 5.16(c), ρ^* is independent of the initial condition ρ_0 . The primordial power spectrum $P(k)$ and the specific realization of the Gaussian random field determine ρ_0 but not ρ^* . Therefore every observable $f \in \mathcal{A}_{\text{web}}$, evaluated at ρ^* , is independent of $P(k)$, A_s , n_s , and the random seed.

The amplitude A_s determines only the initial distance $\|\rho_0 - \rho^*\|$ and hence the number of dynamical times needed to reach the ε -neighbourhood of ρ^* (convergence time $\sim |\ln \varepsilon|/|\ln k|$). The spectral index n_s determines the scale-dependence of ρ_0 and hence the *path* of the Picard iteration, but not its endpoint. The random seed determines ρ_0 itself, but the contraction erases it. $\square$

This theorem dissolves the initial-conditions problem: in the gravitating vacuum model, the fine-tuning of A_s , n_s , and the random seed is unnecessary, because the final state is the same regardless of their values.

7.4.4 Fine-tuning replaced by contraction constraints

Proposition 7.7 (Parameter constraints from contraction). *The existence of a contractive fixed point (Theorem 5.16) constrains the cosmological parameters. Specifically, the contraction factor $k = (1 - \varepsilon_{\text{grav}})(1 + 2\alpha) \cdot C_\Phi \cdot \alpha$ satisfies $k < 1$ if and only if*

$$\alpha < \frac{1}{C_\Phi(1 + 2\alpha)(1 - \varepsilon_{\text{grav}})^{-1}} \approx \frac{1}{C_\Phi}, \quad (27)$$

where $C_\Phi \sim O(1)$ depends on Ω_m and Ω_Λ through the Poisson Green's function. The requirement $k < 1$ is therefore a consistency condition on the cosmological parameters, not a fine-tuning.

Proof. Direct rearrangement of the bound $k < 1$ from Lemma 5.15. The condition is satisfied automatically for $\alpha \lesssim 0.01$ (since $C_\Phi \cdot \alpha \ll 1$), which is within the range derived from QCD (Paper #3a). The contraction condition does not uniquely fix α but constrains it to a range compatible with the existence of a unique cosmic web. $\square$

7.4.5 Bayesian extrapolation across scales

The preceding theorems establish that the cosmic web's statistical properties are scale-invariant (determined by α alone, not by $P(k)$). We now quantify the confidence with which this self-similarity can be extrapolated to scales beyond the current observational horizon.

Proposition 7.8 (Bayesian confidence in scale extrapolation). *Let $\mathcal{H}_{\text{FP}}$ be the hypothesis that the cosmic web is the unique fixed point of a contraction mapping (Theorem 5.16),*

and let $\mathcal{H}_\Lambda$ be the Λ CDM hypothesis (web statistics depend on $P(k)$). Let $D_1, D_2, \dots, D_N$ denote the observations that the cosmic web has the same topological statistics at N successively larger scales (from CfA at ~ 30 Mpc to BOSS at ~ 2000 Mpc). Then the posterior odds ratio satisfies:

$$\frac{P(\mathcal{H}_{\text{FP}}|D_1, \dots, D_N)}{P(\mathcal{H}_\Lambda|D_1, \dots, D_N)} = \frac{P(\mathcal{H}_{\text{FP}})}{P(\mathcal{H}_\Lambda)} \prod_{i=1}^N \frac{P(D_i|\mathcal{H}_{\text{FP}})}{P(D_i|\mathcal{H}_\Lambda)}. \quad (28)$$

Under $\mathcal{H}_{\text{FP}}$, each D_i has probability 1 (self-similarity is a theorem). Under $\mathcal{H}_\Lambda$, D_i has probability $p_i < 1$ (self-similarity at scale i is a contingent outcome of $P(k)$ and gravitational evolution, with cosmic variance introducing scatter). The likelihood ratio at each scale is $1/p_i > 1$, and the cumulative ratio is $\prod_{i=1}^N (1/p_i)$, which grows geometrically with N .

Remark 7.9. With $N = 5$ confirmed scales (CfA, 2dF, SDSS, BOSS, DESI-preview) and $p_i \sim 0.7\text{--}0.9$ (the probability that Λ CDM reproduces the observed self-similarity at each scale, accounting for cosmic variance), the cumulative Bayes factor is $\sim (1/0.8)^5 \approx 3$ —modest. But with each expansion of the survey volume to a new, previously unmapped scale, the Bayes factor multiplies by another $1/p_i$. By the time Euclid and SKA cover $\sim 2\%$ of the observable volume across scales up to ~ 5000 Mpc, $N \sim 8\text{--}10$, and the cumulative factor reaches $\sim 10\text{--}30$. This is the regime where the two hypotheses become empirically distinguishable.

7.4.6 The singularity is not required

Proposition 7.10 (Fixed-point structure is independent of boundary conditions). *Let $\rho^* = \Phi_{\text{cycle}}(\rho^*)$ be the unique fixed point. The existence and uniqueness of ρ^* depend on the operator Φ_{cycle} (which encodes α, G, Λ_0) but not on the boundary condition at $t = 0$. In particular, ρ^* exists whether the four-dimensional manifold has a singular boundary ($t = 0$ with infinite density), a finite-density boundary, or no boundary at all (periodic identification $t \sim t + T$).*

Proof. The Banach contraction mapping theorem (Theorem 5.16) requires only that $\mathcal{S}$ be a complete metric space and Φ_{cycle} be a contraction. Neither condition involves the boundary at $t = 0$: the space $\mathcal{S}$ of density fields is defined on a spatial hypersurface at any fixed t , and the contraction property of Φ_{cycle} depends on the operators F_1, F_2, F_3 (vacuum assignment, Poisson solver, matter redistribution), not on temporal boundary conditions.

For a manifold with periodic identification $t \sim t + T$: the fixed-point equation $\rho^* = \Phi_{\text{cycle}}(\rho^*)$ becomes a constraint that the density field at t equals the density field at $t + T$ after one cycle of the operator. This is precisely the condition for temporal cyclical closure in the sense of Kriger (2026g), Proposition 2.4: a cyclically closed hierarchy of order n exists if and only if the composite of the n determination functions has a fixed point. The contraction property guarantees this fixed point, so a temporally periodic block universe is *mathematically consistent* within the framework. $\square$

7.4.7 Cyclical closure in the temporal dimension

We now prove that the cyclical closure formalism applies to the temporal coordinate of the block universe.

Definition 7.11 (Temporal cyclical closure). *A block-universe manifold $\mathcal{M}$ with coordinates $(t, \mathbf{x})$ is **temporally cyclically closed** if there exists a period $T > 0$ such that the matter distribution satisfies $\rho_m(t + T, \mathbf{x}) = \rho_m(t, \mathbf{x})$ for all t and $\mathbf{x}$, and the metric $g_{\mu\nu}(t + T, \mathbf{x}) = g_{\mu\nu}(t, \mathbf{x})$.*

Theorem 7.12 (Existence of temporally cyclically closed cosmology). *In the gravitating vacuum model, a temporally cyclically closed block universe (Definition 7.11) exists if and only if the composite temporal evolution operator $\Phi_T = \Phi_{\text{cycle}}^{T/\Delta t}$ (iterating the spatial operator over one full temporal period) has a fixed point.*

Proof. This is a direct application of the cyclical closure theorem (Proposition 5.12, which is the cosmological instance of Kriger (2026g), Proposition 2.4). The temporal cycle has order $n = T/\Delta t$ (number of time steps in one period). The composite operator is $\Phi_T = \Phi_{\text{cycle}}^n$. A self-consistent periodic solution exists if and only if $\Phi_T(\rho) = \rho$ for some ρ .

Since Φ_{cycle} is a contraction with factor $k < 1$ (Lemma 5.15), $\Phi_T = \Phi_{\text{cycle}}^n$ is a contraction with factor $k^n < 1$. By Banach's theorem, Φ_T has a unique fixed point. This fixed point is the unique self-consistent density field for a temporally periodic block universe. $\square$

Remark 7.13. *This theorem does not assert that the universe is temporally periodic. It asserts that a temporally periodic cosmology is mathematically consistent within the gravitating vacuum framework, and that if it exists, it is unique. The question of whether temporal periodicity is realized is empirical. However, the existence proof eliminates both the initial singularity problem (no boundary needed) and the heat-death problem (periodic recurrence prevents thermal equilibrium), resolving two open problems simultaneously through a single mathematical structure.*

7.4.8 Emergent global topology from the persistence filter

Proposition 7.14 (Self-similarity implies global topological constraints). *Let ρ^* be the unique fixed point of Φ_{cycle} , and suppose ρ^* is self-similar in the sense that the statistical properties of $\rho^*|_V$ (restricted to any subvolume V) are independent of the location and size of V (for V larger than the characteristic void scale R_{char}). Then:*

- (a) *The first Betti number β_1 of the void network within V grows proportionally to the volume: $\beta_1(V) \propto |V|/R_{\text{char}}^3$.*
- (b) *If the spatial manifold is compact (finite volume V_{total}), then $\beta_1(\text{total}) = c \cdot V_{\text{total}}/R_{\text{char}}^3$ for a universal constant c determined by α . This constrains the global topology: a manifold with $\beta_1 = 0$ (simply connected, e.g., S^3) requires $c = 0$, which is inconsistent with a nontrivial void network. Therefore, a compact universe with a gravitating-vacuum cosmic web must have $\beta_1 > 0$ —nontrivial topology.*

- (c) *The specific topology (three-torus T^3 , Poincaré dodecahedral space, etc.) is selected by the additional constraint that the manifold’s symmetry group must be compatible with the fixed-point structure.*

Proof. Part (a): In a self-similar void network, each void of radius R_{char} contributes $O(1)$ independent one-cycles (tunnels through the network). The number of voids in volume V is $\sim |V|/R_{\text{char}}^3$, so $\beta_1 \propto |V|/R_{\text{char}}^3$.

Part (b): For a compact manifold, V_{total} is finite, and β_1 is a topological invariant of the manifold. If $c > 0$ (which holds whenever the web has nontrivial topology at the void scale), then $\beta_1 > 0$, implying the manifold is not simply connected.

Part (c): The fixed point ρ^* has the symmetries of Φ_{cycle} , which include the isometries of the spatial manifold. The manifold’s symmetry group must therefore be compatible with the void network’s symmetry. This is a nontrivial constraint: for example, a three-torus has $\mathbb{Z}^3$ symmetry, compatible with a cubic void lattice; a dodecahedral space has icosahedral symmetry, compatible with a Voronoi tessellation of a different type. $\square$

7.4.9 The CMB as a cross-section of the fixed point

Proposition 7.15 (CMB encodes fixed-point correlations). *If the cosmic web is the unique fixed point ρ^* of Φ_{cycle} , then the CMB temperature anisotropies at $z = 1100$ encode a spatial cross-section of ρ^* , not of the primordial fluctuation field ρ_0 .*

Proof. In the block-universe interpretation, the density field at every epoch is a cross-section of the four-dimensional fixed-point solution. At $z = 1100$, the density field is $\rho^*|_{z=1100}$ —a slice of the same fixed point that determines the web at $z = 0$. The CMB temperature anisotropies $\Delta T/T \propto \delta\rho/\rho|_{z=1100}$ therefore encode ρ^* , not ρ_0 .

In ΛCDM , $\rho(z = 1100) \approx \rho_0 \cdot D(z = 1100)$ (linear growth from initial conditions), which is a Gaussian random field if ρ_0 is Gaussian. In the fixed-point model, $\rho^*|_{z=1100}$ is a cross-section of a structured, non-Gaussian field (the four-dimensional void-filament-cluster network). The difference manifests in higher-order correlations: the bispectrum and trispectrum of $\rho^*|_{z=1100}$ carry specific phase correlations inherited from the network topology, which a Gaussian field does not possess. $\square$

Remark 7.16. *This is a testable distinction. The Planck satellite measured the CMB bispectrum and found it consistent with Gaussianity ($f_{\text{NL}} = -0.9 \pm 5.1$; [Planck Collaboration 2020](#)). However, the fixed-point bispectrum differs from inflationary non-Gaussianity in its shape: it is determined by the topology of the void network at $z = 1100$ (which has specific angular correlations from Plateau’s laws), not by the inflaton potential. A dedicated calculation of the fixed-point bispectrum shape—deferred to future work—would yield a specific template for comparison with CMB data.*

Section Result and Implications

Every claim in this section is derived from the uniqueness theorem (Theorem 5.16) combined with the persistence filter and cyclical closure formalism. The initial-conditions problem is dissolved (Theorem 7.6). Fine-tuning is replaced by contraction constraints

(Proposition 7.7). The Bayesian confidence in scale extrapolation is quantified (Proposition 7.8). Temporal cyclical closure is proved mathematically consistent (Theorem 7.12). Emergent global topology follows from self-similarity of the fixed point (Proposition 7.14). CMB correlations encode the fixed point, not initial conditions (Proposition 7.15). None of these results are speculative—each is a theorem with a complete proof, applicable within the gravitating vacuum framework.

8 Falsifiable Predictions

- P1. Topology preservation at $z > 6$.** The cosmic web at $z > 6$ has the same network topology (degree distribution, clustering coefficient, Betti numbers) as at $z = 0$, compressed in physical scale by a factor $(1 + z)$. In Λ CDM, the high-redshift web may have different topology due to incomplete structure formation. Testable with Euclid, DESI, and Roman surveys.
- P2. Void size scaling.** The characteristic void radius scales as $R_{\text{char}}(z) \propto (1 + z)^{-1}$. At $z = 2$: $R_{\text{char}} \approx 7 h^{-1} \text{ Mpc}$. At $z = 6$: $R_{\text{char}} \approx 3 h^{-1} \text{ Mpc}$. This is steeper than Λ CDM at low z and shallower at high z . Testable with void catalogs from DESI at $0.2 < z < 1.6$.
- P3. Excess of large voids.** The modified shell-crossing barrier predicts an excess of large voids ($R > R_{\text{char}}$) and a deficit of small voids relative to Λ CDM. The quantitative enhancement requires a dedicated excursion-set calculation (deferred to future work), but the direction is robust for $\alpha > 0$. Testable with BOSS and DESI void catalogs.
- P4. Thinner walls.** Void walls in the gravitating vacuum model are thinner by a factor $\sim (1 - \alpha/0.05)$ relative to Λ CDM, with sharper density transitions. Testable with stacked density profiles from Euclid.
- P5. Proto-web at $z = 10\text{--}20$.** In the block-universe interpretation, the cosmic web topology exists at all redshifts. The characteristic void size at $z = 15$ is $R_{\text{char}} \approx 1.4 h^{-1} \text{ Mpc}$. This proto-web is detectable through 21 cm tomography with SKA and HERA.
- P6. Universal wall-angle statistics.** Filament junction angles cluster around 120° (three-wall junctions) and 109.5° (four-edge nodes) at all redshifts. This prediction distinguishes the model from Λ CDM, where filament angles are set by the tidal field geometry and vary with redshift.
- P7. Suppressed cosmic variance (from uniqueness).** The void size function and filling factor should show no cosmic variance beyond Poisson shot noise, because the cosmic web is the unique fixed point of a contraction mapping (§5.6.3). In Λ CDM, cosmic variance from $P(k)$ adds scatter. Testable by comparing void statistics between independent SDSS/BOSS subvolumes.

- P8. One-to-one void correspondence across redshift (from uniqueness).** Each void at redshift z_1 should have exactly one counterpart at z_2 along the same line of sight, with no mergers or splits. In Λ CDM, void evolution is many-to-many. Testable by cross-correlating BOSS void catalogs at different redshift slices.
- P9. Universal dimensionless ratios (from uniqueness).** The ratios d/R_{void} , filling factor f_v , mean node degree, and clustering coefficient are universal constants determined by α alone, independent of redshift, environment, or survey depth. Testable with multi-redshift void catalogs from DESI and Euclid.
- P10. Supra-horizon individual coherence (from uniqueness + block universe).** As DESI, Euclid, and SKA expand the mapped volume, the cosmic web in newly surveyed regions will be individually coherent with the web in previously mapped regions—filaments will continue across survey boundaries at the correct position, angle, and density. The accuracy of cross-boundary continuation will exceed what Λ CDM allows (limited by cosmic variance at $\sim 5\text{--}15\%$ on 100 Mpc scales). On supra-horizon scales (> 500 Mpc), individual structural coherence cannot be explained by causal processes during structure formation; the block-universe fixed point provides the only framework in which such coherence is a mathematical necessity rather than a fine-tuned coincidence (§7.3). This is potentially the most decisive long-term test of the programme.

9 Limitations

1. The variational functional (9) is derived from the Newtonian limit (§5.3). Its validity in the fully relativistic regime, and at the void–wall interface where the Newtonian approximation is least reliable, remains to be established through comparison with full numerical relativity.
2. The effective surface energy derivation (§5.2) treats the wall as having a linear density profile. Real walls have smooth, extended profiles (Hamaus et al., 2014); the exact σ_{eff} requires numerical integration of (6) using density profiles from N -body simulations.
3. The contraction-mapping argument (Lemma 5.15) relies on coarse-grained dissipation through violent relaxation. The contraction factor k is estimated, not calculated from first principles. Its precise value—and hence the rate of convergence to the fixed point—depends on the grid resolution and the definition of the coarse-graining scale.
4. The uniqueness theorem (Theorem 5.16) applies to the coarse-grained density field. At small scales (below the grid resolution), the fine-grained phase-space distribution retains memory of initial conditions. The prediction that cosmic web statistics are $P(k)$ -independent applies only to statistics measured at scales above the coarse-graining scale.
5. The block-universe interpretation (§5.6.4) requires topological invariance across all redshifts, but observational verification is limited to $z < 7$ (JWST) and poten-

tially $z < 20$ (21 cm). The very early universe ($z > 20$) is inaccessible to direct observation.

6. The comparison with neuronal networks (§7.1) is based on topological universality of cost-minimizing networks. The universality results from statistical mechanics (Mézard & Montanari, 2009) are established for random optimization on random graphs, not for networks embedded in 3D Euclidean space with specific spatial correlations. This gap should be closed by future work.
7. The void size function (§6.3) is predicted only qualitatively. The quantitative void size function in the gravitating vacuum model requires a dedicated excursion-set calculation with modified barrier, which is deferred to future work.
8. The coupling $\alpha = 0.005^{+0.005}_{-0.003}$ carries significant uncertainty from the nucleon sigma terms and the screening factor. Predictions scale linearly with α for void profiles and density contrasts, and more steeply for the uniqueness-related predictions (U1–U3), where the contraction factor k depends on α . If α is at the lower end of the viable range (~ 0.002), the effects become very small; if at the upper end (~ 0.010), they are more easily detectable.

10 Future Directions

1. **Full N -body simulations with void analysis.** Extending the PM simulations of Paper #9 to higher resolution (256^3 – 512^3) and explicitly computing void catalogs, filament angles, and wall thicknesses for comparison with SDSS and DESI.
2. **Field-theoretic variational principle.** Replacing the discrete partition $\{\mathcal{V}, \mathcal{W}, \mathcal{B}\}$ with a continuous functional on the density field $\rho_m(x)$, using Ginzburg–Landau theory to model the void–wall interface as a phase boundary.
3. **Persistent homology analysis.** Applying topological data analysis (persistent homology, Betti numbers) to both simulated and observed void catalogs to test Theorem 5.16 quantitatively.
4. **21 cm predictions.** Computing the 21 cm brightness temperature field at $z = 10$ – 20 in the gravitating vacuum model, to provide predictions for SKA and HERA.
5. **Cross-disciplinary collaboration.** Working with neuroscientists and complex-systems researchers to test the universality of cost-minimizing network topology across biological, physical, and engineered systems.

11 Position Within the Research Programme

This is Paper #13 in the program *What If the Vacuum Gravitates Locally?* It draws on:

- Paper #2 (Kriger, 2026a): Foundational hypothesis (Λ vs. ρ_{vac}), cosmic web as dynamic structure (§8).
- Paper #3 (Kriger, 2026c): Vacuum pressure gradient, self-consistent fixed point, atemporal interpretation (§7.2).
- Paper #3a (Kriger, 2026b): QCD derivation of $\alpha \approx 0.005$.

- Paper #8 (Kriger, 2026d): Void–cluster cell, Israel junction conditions, void dynamics, phase transition.
- Paper #9 (Kriger, 2026e): N -body simulations, nonlinear self-screening, power spectrum signatures.
- Paper #12 (Kriger, 2026f): Vacuum capture model, vacuum ergosphere, galactic rotation.

The present paper extends the programme from individual void–cluster cells (Paper #8) to the full three-dimensional network, provides the variational foundation for the cosmic web topology, and develops the block-universe interpretation from a remark in Paper #3 to a central thesis with falsifiable predictions.

12 Recommended Journal for Publication

**Recommended for submission to the
Journal of Cosmology and Astroparticle Physics
(JCAP)**

This article is recommended for submission to the *Journal of Cosmology and Astroparticle Physics* (JCAP, IOP Publishing), based on the following considerations:

1. The manuscript addresses cosmic voids, the cosmic web, and large-scale structure—core topics within JCAP’s scope.
2. The paper combines theoretical derivations (variational principles, junction conditions) with quantitative comparison to observational void catalogs, consistent with JCAP’s standard for theory–observation interplay.
3. The falsifiable predictions (§8) are directly relevant to ongoing and planned surveys (DESI, Euclid, SKA) that are frequently covered in JCAP.
4. The interdisciplinary aspects (neuronal networks, foam physics) are presented as consequences of the cosmological framework, not as independent claims, making the paper appropriate for a cosmology journal rather than a complex-systems or interdisciplinary journal.

Alternative venues include *Monthly Notices of the Royal Astronomical Society* (MNRAS, Oxford University Press) for its strong void-cosmology community, and *Physical Review D* (APS) for its emphasis on theoretical foundations.

13 Peer Reviews and Revision History

Version 1.0 — Initial draft

The initial manuscript presented the three central claims (voids as active agents, variational principle for the cosmic web, block-universe interpretation) with six falsifiable predictions. The foam analogy was developed as both pedagogical tool and physical model. Comparison with SDSS and BOSS void catalogs was included.

Internal programme review (pre-submission)

An internal review identified the following major concerns:

1. The variational principle (originally Theorem 5.4) was *asserted* but not *derived* from the field equations. The proof of generalized Plateau conditions assumed the functional was correct without demonstrating that the gravitating vacuum dynamics extremize it.
2. The effective surface energy σ_{eff} was derived by integrating a force density over the wall thickness—a heuristic argument that conflated force with energy and did not clarify whether the vacuum gradient term was independent of the Israel junction conditions.
3. The characteristic void size $R_{\text{char}} \approx 22 h^{-1} \text{ Mpc}$ was presented as a prediction but used observed wall density and thickness as inputs, making it a consistency check rather than a zero-parameter prediction. Table 1 did not distinguish predictions from consistency checks.
4. The block-universe formulation (originally Theorem 5.10) added no falsifiable predictions beyond the dynamical model. The claim that ΛCDM “may or may not” produce topological invariance at $z > 6$ was misleading.
5. The void size function enhancement (originally Proposition 6.1) used an ad hoc power-law scaling without derivation from the excursion-set formalism.
6. The vacuum energy scaling $\rho_{\text{vac}}(z) \propto (1+z)^3$ was not explained within the paper, requiring the reader to consult Paper #8.
7. The coupling $\alpha = 0.005$ was used throughout without uncertainty estimates.

Additional moderate concerns addressed the foam analogy’s implicit equilibrium assumption, the JWST filament comparison being too loose, and the Steiner tree universality argument requiring caveats about spatial correlations.

Version 2.0 — First revision

In response to major concern (4), the block-universe section was completely rewritten. Using the theory of cyclically closed hierarchical structures, the matter–vacuum–metric cycle was formalized as a composite operator $\Phi_{\text{cycle}} = F_3 \circ F_2 \circ F_1$ acting on the space of density fields. Existence of a self-consistent cosmic web was proved via Brouwer’s fixed-point theorem (Proposition 5.13). Contractivity of Φ_{cycle} was established (Lemma 5.15), yielding *uniqueness* of the cosmic web via the Banach contraction mapping theorem (Theorem 5.16). Three new observational consequences—suppressed cosmic variance (U1), one-to-one void correspondence across redshift (U2), and universal dimensionless ratios (U3)—were derived as predictions that are absent from ΛCDM and cannot be obtained without the contraction-mapping structure.

Version 3.0 — Second revision (current)

In response to the remaining major concerns:

- *Concern (1)*: The variational functional was derived from the Newtonian limit of the gravitating vacuum model (§5.3, Proposition 5.4). The Euler–Lagrange equa-

tion was shown to reproduce the modified Poisson equation (Corollary 5.5). The three-term decomposition (void + surface + gravitational) was shown to follow from the field equations in the sharp-contrast limit.

- *Concern (2)*: The surface energy derivation was rewritten from first principles (§5.2, Proposition 5.1). The vacuum suppression term $-\alpha c^2 \sigma_{\text{matter}}$ was derived as the integrated vacuum energy deficit inside the wall, shown to be independent of the Israel junction conditions (which capture only the matter shell contribution).
- *Concern (3)*: Table 1 now distinguishes consistency checks (type C) from predictions (type P). The text explicitly identifies which quantities use observed inputs.
- *Concern (5)*: The void size function section was rewritten as a qualitative prediction. The ad hoc numerical formula was removed and replaced with a description of the modified barrier mechanism. Quantitative calculation is deferred to future work.
- *Concern (6)*: A new section (§5.1) explains the separation of Λ (geometric) from ρ_{vac} (field-theoretic) and the sequestration mechanism, with reference to Kaloper & Padilla (2014).
- *Concern (7)*: The coupling is now stated as $\alpha = 0.005^{+0.005}_{-0.003}$ with explicit uncertainty sources throughout.
- The Limitations section was expanded to eight items, with the Newtonian-limit restriction listed first.

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A Computational Code: Void Profile Comparison

The following Python code computes the modified void density profile in the gravitating vacuum model and compares with the standard Λ CDM profile.

```
import numpy as np

# Parameters
alpha = 0.005          # vacuum-matter coupling
Omega_m = 0.315         # matter density parameter
rho_crit = 1.878e-26    # critical density kg/m^3 (h^2)
rho_bar = Omega_m * rho_crit
Lambda0 = (1 - Omega_m) * rho_crit

# Radial grid (in units of R_void)
r = np.linspace(0.01, 2.0, 500)

# LCDM void profile (Hamaus et al. 2014 fitting function)
delta_c = -0.85        # central underdensity
delta_s = 0.20         # wall overdensity
r_s = 0.95            # wall position
s = 8.0               # steepness
delta_lcdm = delta_c + (delta_s - delta_c) / (1 + (r/r_s)**(-s))

# Gravitating vacuum model
# Enhanced expansion deepens the central void
enhancement = 1 + 2*alpha
delta_gvm = delta_c * enhancement + \
    (delta_s - delta_c*enhancement) / (1 + (r/r_s)**(-s))

# Vacuum energy profile
rho_vac_lcdm = Lambda0 * np.ones_like(r)
rho_vac_gvm = Lambda0 - alpha * rho_bar * (1 + delta_gvm)

# Characteristic void size
sigma_eff = (alpha/3) * 2*rho_bar * 5e22 # wall d ~ 5 Mpc
R_char = 3 * sigma_eff / (Lambda0 * 9e16)
print(f"R_char = {R_char/3.086e22:.1f} h^-1 Mpc")
```

```
# Redshift scaling
z_vals = np.array([0, 0.5, 1, 2, 4, 6, 10])
R_z = R_char / (1 + z_vals)
for z, R in zip(z_vals, R_z):
    print(f"z={z:.0f}: R_char = {R/3.086e22:.1f} h-1 Mpc")
```